Moltbook Session — Supporting Notes

Internal Documentation, Verification & Model Audit

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    - 8. Implementation Roadmap
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    - 10. Honest Assessment: What V2 Achieves and Doesn't
    - 11. Immediate Next Steps (No Code Yet)
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    - 1. Event Definitions (Exact)
    - 2. Mathematical Specification (Exact)
    - 3. Parameter Strategy
    - 4. Calibration Plan
    - 5. Validation Plan
    - 6. Risks and Open Questions
    - 7. Deliverables and Next Steps
    - 8. Explicit Absence

Documentation Index

This directory contains documentation for the Moltbook session repository.

For General Readers (Public-Facing)

Start with the main content: - `../content/README.md` — Overview of all content chapters - `../content/01-intro.md` — Introduction and core thesis - `../README.md` — Project landing page

Internal Documentation

The following are for verification and quality assurance:

Verification Reports (Internal)

- `verification/VERIFICATION_REPORT.md` — Repo verification
- `verification/CLAIM_AUDIT.md` — Claim-by-claim audit
- `verification/ANALYSIS_AUDIT.md` — Analysis validation
- `verification/SLIDE_SYSTEM_DECISION.md` — Technical decision log

QA Reports (Internal)

- `qa/FINAL_QA_REPORT.md` — Final QA pass results
- `qa/PRESENTATION_REDESIGN_REPORT.md` — Design evolution
- `qa/FINAL_PRESENTATION_POLISH_REPORT.md` — Polish pass log

Model Audit (Internal — Developers/Reviewers Only)

- `model_audit/README.md` — Entry point for model audit docs

Important: The `model_audit/` directory contains extensive technical documentation about the forecasting model's design, validation, and limitations. These are **internal working documents** for transparency and reproducibility, not part of the public-facing project narrative.

Release Information

- `PUBLIC_RELEASE_REPORT.md` — Release cleanup log

Navigation tip: If you're reading this project for the first time, start with `../content/01-intro.md` rather than these internal docs.

Part 2: Release Report

Public Release Report

Executive Summary

This pass turned the repository from an internal working deck project into a more portable public release. The build path is still reproducible with `uv` and `bun`, but machine-specific links, root clutter, and accidental artifacts were removed.

What Was Reorganized

- moved verification reports into `docs/verification/`
- moved the final QA report into `docs/qa/`
- moved preview renders into `docs/qa/previews/`
- moved the generated deck into `release/`
- added `docs/README.md` and `docs/verification/README.md` as navigation entry points

Why The New Structure Is Better

- the repo root now highlights source, build config, and the release artifact instead of audit clutter
- verification and QA material are grouped under one `docs/` surface
- the final deck has an intentional home in `release/`
- portable relative links now work on GitHub instead of depending on one workstation path

What Was Removed

- `Document1.docx`
- `Presentation1.pptx`

Both files were tracked Office artifacts with no references in the repo and no value to the public release.

What Was Kept And Why

- `release/Moltbook.pptx`
  - kept as the main downloadable artifact for public visitors
- `docs/qa/previews/*`
  - kept because they document slide-by-slide QA without requiring Office
- `requirements.txt`
  - kept as a pip fallback even though `uv` plus `pyproject.toml` and `uv.lock` are the preferred path

Public-Facing Issues Fixed

- removed absolute local paths from docs and links
- replaced machine-specific `cd /home/...` commands with repo-root commands
- moved public reports out of the root
- made deck output and preview output locations explicit

- added ignore coverage for LibreOffice lock debris

Remaining Tradeoffs

1. The repo still does not declare an open-source license. That is a publication decision rather than a technical one.
2. Preview renders are useful QA evidence, but they are not a perfect substitute for opening the deck in PowerPoint or LibreOffice.
3. `requirements.txt` duplicates dependency intent already captured in `pyproject.toml` and `uv.lock`, but it remains useful for readers who do not use `uv`.

Exact Commands Run

```
mkdir -p docs/verification docs/qa release
mv VERIFICATION_REPORT.md docs/verification/VERIFICATION_REPORT.md
mv CLAIM_AUDIT.md docs/verification/CLAIM_AUDIT.md
mv ANALYSIS_AUDIT.md docs/verification/ANALYSIS_AUDIT.md
mv SLIDE_SYSTEM_DECISION.md docs/verification/SLIDE_SYSTEM_DECISION.md
mv FINAL_QA_REPORT.md docs/qa/FINAL_QA_REPORT.md
mv Moltbook.pptx release/Moltbook.pptx
rm -f Document1.docx Presentation1.pptx
mkdir -p docs/qa/previews
mv qa/previews/* docs/qa/previews/
rmdir qa/previews qa
MPLCONFIGDIR=/tmp/matplotlib UV_CACHE_DIR=.uv-cache uv run analyses/token_usage.py
MPLCONFIGDIR=/tmp/matplotlib UV_CACHE_DIR=.uv-cache uv run analyses/ai_trends.py
MPLCONFIGDIR=/tmp/matplotlib UV_CACHE_DIR=.uv-cache uv run analyses/forecast_model.py
BUN_TMPDIR=/tmp BUN_INSTALL_CACHE_DIR=.bun-cache bun run build:deck
UV_CACHE_DIR=.uv-cache uv run scripts/export_slide_previews.py
```

Clean-Worktree Proof

At handoff, the intended proof standard is:

- the moved paths are committed
- the rebuilt deck exists in `release/`
- preview renders exist in `docs/qa/previews/`
- `git status --short` is clean

Part 3: Verification Reports

Verification Report

Executive Summary

This repo did not start in a verification-ready state.

The main issues were:

- the AI trends chart was generated from synthetic data that looked empirical
- the documented analysis commands failed from the repo root
- the markdown deck overstated confidence on several points
- the original MiniMax M2.7 vs Claude Opus 4.6 comparison was not source-aligned
- the repo mixed multiple incompatible deck artifacts

The repository is in a materially stronger state:

- all three analysis scripts run reproducibly with `uv`
- all generated figures save correctly from the repo root
- the trends chart is source-anchored instead of synthetic
- the token-cost figure is explicitly labeled as an illustrative scenario
- the forecast model is explicitly labeled as assumption-driven
- a native `.pptx` build path now exists via `bun` + `PptxGenJS`
- the slide outline and content notes were rewritten to match the verified framing

Repo Truth vs Prior Claims

What was true

- Moltbook does market itself as a social network for AI agents.
- Moltbook's Terms do deny legal eligibility to AI agents and keep responsibility with the human owner.
- OpenClaw does document context loading, agent routing, and per-agent sandbox/tool policies.
- The repo did already contain working Monte Carlo logic and a useful cost-scenario idea.

What was weak or wrong

- `analyses/ai_trends.py` fabricated ECI and saturation curves rather than plotting source data.
- The old deck treated the `$0.38 per social cycle` number as if it were observed rather than assumed.
- The old deck's MiniMax M2.7 vs Claude Opus 4.6 claim was too precise and could not be validated from the repo's then-current sources.
- The old scripts wrote to `../assets/...`, which broke the documented root-level commands.
- The repo contained deck artifacts that were not aligned with the markdown deck.

What Was Verified

- Moltbook homepage language and terms text
- OpenClaw context and multi-agent documentation
- Stanford HAI summary metrics used in the talk

- Epoch methodology and trend snapshot framing
- the arithmetic behind the token-cost scenario
- the Monte Carlo output from the forecast model

What Was Corrected

- analysis scripts now load explicit inputs from `data/*.json`
- figures now save relative to the repo root
- the trends chart now uses sourced metrics only
- the MiniMax section now uses official MiniMax and Anthropic pages with narrower wording
- the token-cost section now distinguishes documented anchors from assumptions
- the forecast model now reports threshold and floor sensitivity
- the deck now uses safer wording around trend curves and autonomy claims
- a native PowerPoint build path was implemented in scripts/build_deck.ts

Biggest Credibility Risks Found

1. Synthetic data presented as if it were sourced.
2. Source mismatch in the original MiniMax comparison.
3. Over-strong wording around “AI-only” and social autonomy.
4. Forecast outputs that looked more precise than the assumptions justified.
5. Broken reproduction instructions.

Highest-Value Improvements Made

1. Replaced the synthetic trends visual with a source-anchored one.
2. Moved assumptions into explicit data files.
3. Rebuilt the deck as a native `.pptx`.
4. Rewrote the talk around what can actually be defended.
5. Added audit documents so remaining uncertainty is visible.

Evidence Snapshots

OpenClaw context anchor

OpenClaw context screenshot
OpenClaw context screenshot

Moltbook terms anchor

Moltbook terms screenshot
Moltbook terms screenshot

Rebuilt token-cost figure

Token breakdown
Token breakdown

Rebuilt trends figure

AI trends
AI trends

Rebuilt forecast figure

Forecast distribution
Forecast distribution

Remaining Uncertainties

- The Meta acquisition claim exists in reputable reporting, but I did not find a first-party Meta announcement. The deck therefore avoids calling it “official” in first-party terms.
- The Tsinghua Moltbook paper is strong evidence, but it is still a recent paper and should be treated as high-value analysis rather than settled canon.
- The forecast model remains assumption-driven. It is now explicit about that, but it is still not empirical forecasting.
- The token-cost scenario remains illustrative rather than observed.

Reproduction Result

Successful with:

- `UV_CACHE_DIR=.uv-cache uv sync`
- `MPLCONFIGDIR=/tmp/matplotlib UV_CACHE_DIR=.uv-cache uv run analyses/token_usage.py`
- `MPLCONFIGDIR=/tmp/matplotlib UV_CACHE_DIR=.uv-cache uv run analyses/ai_trends.py`
- `MPLCONFIGDIR=/tmp/matplotlib UV_CACHE_DIR=.uv-cache uv run analyses/forecast_model.py`
- `BUN_TMPDIR=/tmp BUN_INSTALL_CACHE_DIR=.bun-cache bun install`
- `BUN_TMPDIR=/tmp BUN_INSTALL_CACHE_DIR=.bun-cache bun run build:deck`

Final Standard

The repo is now much closer to this standard:

- every important number is traceable to code and either a source or an explicit assumption
- every generated chart is reproducible
- the deck is stronger and more honest than before
- the slide pipeline is justified and upgraded

Claim Audit

Claim text	Location	Source(s)	Verification status	Reproduction status	Action taken	Notes
Moltbook is “A Social Network for AI Agents” and humans may observe	Slide 2, `content/01-intro.md`	Moltbook homepage	Verified	Source-supported, not code-derived	Kept	Directly supported by homepage copy
Moltbook Terms deny legal eligibility to AI agents and keep responsibility with the human owner	Slide 2, `content/04-kritiek.md`	Moltbook Terms of Service	Verified	Source-supported, not code-derived	Kept	Strong primary-source support
OpenClaw treats context as everything sent to the model for a run	Slide 3, `content/03-agents-md.md`	OpenClaw docs	Verified	Source-supported, not code-derived	Kept	Core architectural framing
Multi-agent OpenClaw setups use per-agent sandbox and tool policies	Slide 3, `content/02-ai-agents.md`	OpenClaw multi-agent docs	Verified	Source-supported, not code-derived	Kept	Good support for “agent network” system framing
One social cycle costs about $0.38 on Claude Opus 4.6	Slide 4, `content/03-agents-md.md`	OpenClaw docs anchor, Anthropic pricing, repo assumptions	Reframed	Reproduced from code but assumption-driven	Narrowed	Now explicitly labeled illustrative, not measured
Moltbook identity claims are shaky because author activity is mixed across autonomous-	Slide 5, `content/04-kritiek.md`	Tsinghua paper	Verified	Source-supported, not code-derived	Kept	Replaced rounded repo percentages with paper values

Claim text	Location	Source(s)	Verification status	Reproduction status	Action taken	Notes
leaning, human-leaning, and ambiguous categories						
Frontier capability and efficiency trends are moving fast	Slide 6, `content/05-trends.md`	Stanford HAI, Epoch	Verified	Reproduced from source-backed data file	Kept	Strongest trend claim preserved
Stanford HAI reports >280x decline in cost to reach GPT-3.5-level quality between Nov 2022 and Oct 2024	Slide 6, `content/05-trends.md`	Stanford HAI	Verified	Reproduced from normalized endpoints in code	Kept	Now visualized as a normalized derived curve
ECI uses exactly 37 benchmarks	Old slides and notes	Epoch ECI page	Ambiguous / source mismatch	Not directly reproduced	Rewritten	Deck now says “dozens of benchmarks” because Epoch page is internally inconsistent
MiniMax M2.7 is much cheaper than Claude Opus 4.6 by official list price and still competitive on some agentic/coding tasks	Slide 6-7, `content/05-trends.md`	MiniMax official model page, MiniMax official pricing docs, Anthropic official Opus 4.6 page	Verified but narrow	Source-supported, not locally benchmark-reproduced	Reintroduced with tighter wording	Safe summary is economic plus limited competitiveness; “basically equal to Opus” remains unsupported
The forecast median is 2038 in the base case	Slide 8, `content/06-forecast.md`	Repo forecast code	Verified	Reproduced from code	Kept with caveat	Now explicitly labeled assumption-driven
Threshold changes alone move	Old implied framing	Repo forecast model	Not supported in current parameterization	Reproduced and contradicted	Rewritten	Floors bind before the headline

Claim text	Location	Source(s)	Verification status	Reproduction status	Action taken	Notes
the forecast substantially						threshold in current setup
Moltbook proves autonomous digital society is already here	Old rhetorical framing	None	Unsupported	Not applicable	Removed	Replaced with narrower thesis

Analysis Audit

`analyses/token_usage.py`

Input source

- data/token_usage_assumptions.json
- OpenClaw docs anchor for context overhead
- Anthropic pricing
- Artificial Analysis pricing for MiniMax-M2.5

Method summary

- load documented context anchor and explicit scenario assumptions
- compute one illustrative read-reply-post cycle
- compute per-model token costs
- generate scale examples

Assumptions

- scenario is illustrative, not measured from Moltbook logs
- AGENTS/project instructions, persona, and prior history token counts are explicit assumptions

Validation result

- arithmetic reproduced successfully
- script now runs from repo root
- figure now writes to `assets/token_breakdown.png`

Issues found

- prior script saved to the wrong relative path
- prior narrative did not clearly distinguish measurement from scenario arithmetic

Fixes made

- moved assumptions into JSON
- fixed save paths
- rewrote titles and labels to say “illustrative” and “assumption-driven”

Does the chart support the slide claim?

Yes, after the claim was narrowed to: an illustrative social cycle under stated assumptions costs about `$0.377` on Claude Opus 4.6.

`analyses/ai_trends.py`

Input source

- data/ai_trends_metrics.json
- Stanford HAI summary metrics
- Epoch AI methodology and homepage trend snapshot
- MiniMax official M2.7 model and pricing pages
- Anthropic official Opus 4.6 page

Method summary

- plot only explicit sourced metrics
- derive a normalized cost-decline line from sourced endpoints
- include an audit note panel instead of synthetic saturation curves

Assumptions

- the cost-decline line is a normalization from sourced endpoints, not a digitized original dataset
- the MiniMax/Opus section uses official vendor-reported numbers rather than a neutral matched evaluation sheet

Validation result

- script runs successfully from repo root
- figure now saves to `assets/ai_trends.png`

Issues found

- prior script fabricated ECI scores and saturation curves
- prior script embedded a too-confident model comparison
- the first verification pass was too conservative on M2.7 because official MiniMax documentation had not yet been incorporated

Fixes made

- removed synthetic time series
- replaced exact ECI benchmark count with an internal-source inconsistency note
- rewrote comparison framing around what can actually be defended
- upgraded MiniMax from “unsupported” to a narrow official-source-backed economics/competitiveness claim

Does the chart support the slide claim?

Yes, for the narrowed claim that the strongest defendable trends are in cost, compute, efficiency, broad capability context, and that MiniMax M2.7 shows a much clearer price advantage than quality gap.

`analyses/forecast_model.py`

Input source

- data/forecast_scenarios.json

Method summary

- simulate seven readiness pillars under three scenarios

- compute a weighted geometric readiness index
- apply threshold and floor rules
- report crossing-year probabilities
- run threshold and floor sensitivity checks

Assumptions

- all growth, volatility, and shock parameters are assumptions
- weights, floors, and threshold are scenario design choices

Validation result

- script runs successfully from repo root
- outputs reproduced with explicit seed `42`
- chart saves to `assets/forecast_distribution.png`

Issues found

- prior script saved to the wrong path
- prior presentation framing made the output look too point-estimate-like
- threshold sensitivity alone did not move results because floor constraints bound first

Fixes made

- moved all scenario inputs into JSON
- fixed save paths
- added floor sensitivity
- rewrote printed interpretation to emphasize assumption error over simulation error

Does the chart support the slide claim?

Yes, for the narrowed claim that the model is useful as a scenario discipline and that the current base case crosses with about `28.1%` probability by 2040 under the stated assumptions.

Slide System Decision

Decision

Use **PptxGenJS** as the production slide pipeline.

Keep Markdown only as a human-readable outline and speaker-note aid.

Why the old workflow lost

The old repo state effectively treated Markdown as the main slide source, but the actual binary deck artifacts were already drifting away from the Markdown.

That failed on the key criteria:

- weak evidence of deterministic deck generation
- poor alignment between source and output
- hard to enforce citations, notes, and layout consistency
- no clear native editable PowerPoint path

Comparison Matrix

Option	Reproducibility	Visual control	Native `.pptx`	Notes support	Deterministic assets	Portability	Fit for this repo
Legacy Markdown workflow	Weak in current repo	Medium	No	Varies	Weak	Medium	Lost because source/output drift already happened
PptxGenJS	Strong	Strong	Yes	Yes	Strong	Strong for conference decks	Best fit
reveal.js	Strong	Strong for web	No	Presenter notes yes	Strong	Strong for web talks	Good alternative, not best here
Slidev / Marp	Medium to strong	Medium	Usually export-oriented, not native	Medium	Strong	Good for markdown-first teams	Easier, but lower control for this talk

Why PptxGenJS Wins

- native editable `.pptx` output
- deterministic build from code
- good support for images, text, tables, and speaker notes

- easier to keep citations and footer text consistent
- better fit for a branded conference deck than a markdown-first workflow

Implemented Result

- added package.json
- added `bun` lockfile
- added scripts/build_deck.ts
- `bun run build:deck` now rebuilds release/Moltbook.pptx

Build Commands

```
BUN_TMPDIR=/tmp BUN_INSTALL_CACHE_DIR=.bun-cache bun install
BUN_TMPDIR=/tmp BUN_INSTALL_CACHE_DIR=.bun-cache bun run build:deck
```

Markdown's New Role

Markdown still has value here, but a narrower value:

- outline and speaker-note source
- repo-readable narrative draft
- audit-friendly text review

It is no longer the production deck system.

Part 4: QA Reports

Final QA Report

Executive Summary

This was a final pre-talk QA pass on the native release/Moltbook.pptx, not another broad rewrite.

The deck now clears the remaining high-risk issues from the prior verification pass:

- the MiniMax section has been upgraded from “remove entirely” to a narrow official-source-backed claim
- the trend and forecast slides now carry visible methodological disclaimers on-slide
- the native `.pptx` was rebuilt and inspected slide by slide with generated preview renders
- the most analytically risky residual phrasing was tightened again

What Changed In This Pass

0. The native deck now uses a reusable BoostMeUp housestyle

Updated files:

- scripts/pptx-brand.ts
- scripts/build_deck.ts
- assets/brand/boostmeup-logo.png
- assets/brand/boostmeup-mark.png

Applied housestyle:

- dark background: `#12192c`
- light background: `#ffffff`
- red accent: `#e93325`
- gold accent: `#f2ad18`

The branding layer is now reusable for future `PptxGenJS` decks instead of being embedded directly in one build script.

1. MiniMax M2.7 was reintroduced carefully

Updated files:

- data/ai_trends_metrics.json
- analyses/ai_trends.py
- content/05-trends.md
- scripts/build_deck.ts
- CLAIM_AUDIT.md

New safe deck position:

- official pricing clearly supports a major economics gap
- official vendor pages support “still competitive on some agentic/coding tasks”
- they do **not** support “basically equal to Opus 4.6”

Current safe line:

> The price gap is clearer than the quality gap.

2. Visible methodology disclaimers were added to the deck itself

Changed slides:

- Slide 6 now carries a visible trend-method note
- Slide 8 now carries a visible forecast-method note

This matters because the caution no longer depends on speaker notes alone.

3. Closing structure was tightened

The closing slide now uses stronger panel structure to read more like a presentation slide and less like a sparse source draft.

QA Method

Build validation

Used the repo's current supported workflow:

```
MPLCONFIGDIR=/tmp/matplotlib UV_CACHE_DIR=.uv-cache uv run analyses/ai_trends.py
BUN_TMPDIR=/tmp BUN_INSTALL_CACHE_DIR=.bun-cache bun run build:deck
UV_CACHE_DIR=.uv-cache uv run scripts/export_slide_previews.py
```

Slide inspection method

PowerPoint or LibreOffice was not available during this QA pass.

So the visual QA used:

1. the final generated `.pptx`
2. direct XML inspection of slide text in the built deck
3. low-fidelity PNG preview renders generated from the built `.pptx` via scripts/export_slide_previews.py

That is sufficient for:

- title-length risk
- spacing consistency
- image placement
- chart legibility
- footer/disclaimer presence
- slide-by-slide note presence

It is not a pixel-identical Office renderer, so that residual limitation is stated explicitly.

Slide-By-Slide QA Result

Preview metadata: docs/qa/previews/preview-metadata.json

- Slide count inspected: `10`
- Slides with notes detected: `10`

Outcome table

Slide	Focus	QA result	Action
1	Title / opening hierarchy	Accept	No change after final pass
2	Primary-source framing	Accept	No change
3	OpenClaw architecture framing	Accept	No change
4	Token-cost caveat	Accept	Shortened title for lower fit risk
5	Identity / governance caution	Accept	No change
6	Trends	Improved	Shorter title, visible method note, narrower MiniMax framing
7	MiniMax rhetoric risk	Improved	Rewritten around safe vs unsafe language
8	Forecast	Improved	Shorter title, visible method note
9	Synthesis	Accept	No change
10	Closing	Improved	Added panel structure for stronger hierarchy

Inline Evidence

Contact sheet from the generated deck

Deck contact sheet
Deck contact sheet

Slide 6: trend slide after MiniMax tightening and visible method note

Slide 6 preview
Slide 6 preview

What changed:

- visible trend-method disclaimer
- MiniMax claim narrowed to official-source-backed economics + competitiveness
- no return to “basically equal to Opus” rhetoric

Slide 7: explicit MiniMax caution slide

Slide 7 preview
Slide 7 preview

What changed:

- left panel says what is fair to say
- right panel says what still needs caution
- the Sonnet 4.6 vs Opus 4.6 nuance is now explicit

Slide 8: forecast slide with on-slide methodology disclaimer

Slide 8 preview
Slide 8 preview

What changed:

- forecast now visibly reads as scenario discipline, not prophecy
- disclaimer survived into the slide body area instead of hiding in notes

Slide 10: closing slide with improved structure

Slide 10 preview
Slide 10 preview

What changed:

- stronger panel structure
- clearer separation between closing thesis and discussion prompts

Remaining Credibility Risks

1. The MiniMax/Opus comparison is still vendor-reported, not a single independent matched eval sheet.
2. The forecast remains assumption-driven.
3. The token-cost figure remains an illustrative scenario, not an observed Moltbook trace.
4. Native Office rendering was not available during this QA pass, so the visual check used generated previews rather than PowerPoint itself.

None of those block the deck from presentation use if they are spoken honestly.

Final Verdict

The deck is now suitable for real presentation use.

Notable reasons:

- the strongest remaining analytical risk has been narrowed, not hidden
- the methodology caveats are visible on-slide
- the deck build remains deterministic via `bun`
- the analyses remain reproducible via `uv`

Files Changed In This Pass

- data/ai_trends_metrics.json

- analyses/ai_trends.py
- content/05-trends.md
- scripts/build_deck.ts
- scripts/export_slide_previews.py
- slides/slides-main.md
- CLAIM_AUDIT.md
- ANALYSIS_AUDIT.md
- VERIFICATION_REPORT.md
- assets/ai_trends.png
- release/Moltbook.pptx

Clean-Worktree Check

This report is complete only when:

- the deck has been rebuilt
- preview renders have been regenerated
- changes are committed
- `git status --short` is clean

Presentation Redesign Report

Executive Summary

This pass turned the verified Moltbook deck into a stronger Dutch live presentation. The analytical posture stayed narrow and source-safe, but the slide surface moved away from report-like layouts toward clearer keynote roles, stronger rhythm, and less on-slide text.

Design Diagnosis Of The Old Deck

The previous deck had three structural design problems:

1. Slides 2–8 still felt like a verified research document poured into slide boxes.
2. Too many slides relied on the same pattern: title, medium screenshot, medium text block.
3. The presenter-facing language stayed too English and too audit-shaped for a Dutch live talk.

The strongest existing direction was slide 1: darker, more branded, and more event-like. That became the anchor for the redesign.

What Changed Structurally

- rebuilt slides 2–8 around clearer presentation jobs instead of repeating one layout pattern
- moved several slides from screenshot-led layouts to claim / meaning / caution compositions
- reduced visible text and shifted nuance back into speaker notes
- made dark slides carry synthesis, caution, and closing emphasis
- used light slides where data or contrast benefited from it

What Changed Visually

- extended the BoostMeUp dark keynote language beyond the title slide
- added stronger cards, stat blocks, takeaway bands, and pill labels
- removed the generic “grey document panel” feel as the default pattern
- simplified the trend visual into a more presentation-readable chart
- improved the preview renderer so QA can reflect dark slide backgrounds more honestly

What Changed Linguistically For Dutch Delivery

- translated the slide surface into Dutch
- removed or reduced English audit phrasing on the slide surface
- tightened titles into shorter spoken-stage language
- kept English only where the original source phrase itself matters

Important preserved line:

> De prijskloof is duidelijker dan de kwaliteitskloof.

That keeps the exact meaning of the verified English framing.

Slide-By-Slide Improvements

Slide 1

- kept the dark branded anchor
- tightened the title into a stronger Dutch opening
- made the thesis more stage-like and less repo-like

Slide 2

- replaced “source screenshot plus explanation” with a clearer claims-versus-limits contrast
- made the legal tension visually explicit

Slide 3

- replaced documentation-dump energy with a simplified architectural reading
- kept the source anchor in notes and with a small visual reference

Slide 4

- made scenario-versus-measurement explicit through layout, not just caveat text
- foregrounded the context-reload burden instead of a dense multi-panel chart

Slide 5

- turned autonomy uncertainty into memorable stat cards
- made the conclusion legible in seconds

Slide 6

- simplified the trend figure for live readability
- pushed the cautious interpretation into a clear side panel and takeaway band

Slide 7

- turned the MiniMax slide into a crisp price-versus-quality framing
- preserved caution without making the slide defensive

Slide 8

- reduced spreadsheet energy
- kept the forecast as a scenario tool with visible methodological restraint
- emphasized floor constraints as the real model lesson

Slide 9

- sharpened the synthesis into four bottleneck cards
- made the payoff more keynote-like

Slide 10

- strengthened the Dutch closing thesis
- demoted discussion prompts beneath the final takeaway

Tradeoffs Between Punch And Caution

1. The trend slide stays more restrained than a hype deck because bounded-benchmark caution remains visible.
2. The MiniMax slide foregrounds price and keeps quality caveats explicit rather than chasing a harder comparison headline.
3. The forecast slide became less data-dense on purpose so the audience remembers the methodological lesson instead of a pseudo-precise date.

Files Changed

- scripts/build_deck.ts
- scripts/export_slide_previews.py
- analyses/ai_trends.py
- slides/slides-main.md
- assets/ai_trends.png
- release/Moltbook.pptx
- docs/qa/previews/contact-sheet.png

Exact Commands Run

```
MPLCONFIGDIR=/tmp/matplotlib UV_CACHE_DIR=.uv-cache uv run analyses/ai_trends.py
BUN_TMPDIR=/tmp BUN_INSTALL_CACHE_DIR=.bun-cache bun run build:deck
UV_CACHE_DIR=.uv-cache uv run scripts/export_slide_previews.py
```

Clean-Worktree Proof

At handoff, the intended proof standard is:

- the deck has been rebuilt
- previews have been regenerated
- the redesign changes are committed
- `git status --short` is clean

Final Presentation Polish Report

Executive Summary

This pass tightened the Dutch keynote deck from “good verified talk” to a stronger live presentation. The main work was not new research. It was reducing slide density, making the one takeaway per slide more dominant, and improving the payoff on the evidence, synthesis, and closing slides.

Biggest Remaining Weaknesses Found

The prior Dutch deck was already materially better, but it still had these weaknesses:

1. Slides 3, 4, 6, and 8 still carried too much simultaneous information.
2. The strongest takeaway did not always win visually within two seconds.
3. Synthesis and closing still landed more as correct conclusions than as payoff moments.
4. Some evidence slides still leaned too much on “showing the work” instead of staging the point.

What Changed Visually

- reduced evidence density on the most technical slides
- made cost, trend, and forecast takeaways more dominant
- pushed more secondary detail into lower-priority visual zones or notes
- strengthened the dark-slide rhythm for caution, synthesis, and payoff
- kept the deck inside one coherent BoostMeUp visual system

What Changed Structurally

Slide 3

- simplified the architecture read into a cleaner system framing
- reduced the conceptual sprawl so “system, not organism” lands faster

Slide 4

- reduced the cost slide to one dominant comparison
- clarified scenario-versus-measurement more aggressively

Slide 6

- shifted the trends slide toward a stronger top-line thesis
- made economics and infrastructure the visual winners over chart clutter

Slide 8

- made the floor insight the primary memory object
- pushed raw forecast numbers into secondary support

Slide 9

- strengthened the synthesis payoff with a more explicit takeaway

Slide 10

- sharpened the final rhetorical landing
- moved discussion prompts out of the main slide surface

Which Slides Improved Most

Most improved in this pass:

1. Slide 4: costs
2. Slide 6: trends
3. Slide 8: forecast
4. Slide 10: closing

What Was Cut Or Simplified

- reduced multi-point evidence competition on the cost slide
- reduced the role of raw forecast numbers on the forecast slide
- reduced visible discussion material on the closing slide
- reduced the need to read multiple separate claims before understanding the trend slide

Files Changed

- scripts/build_deck.ts
- slides/slides-main.md
- release/Moltbook.pptx
- docs/qa/previews/contact-sheet.png
- docs/qa/previews/preview-metadata.json

Exact Commands Run

```
BUN_TMPDIR=/tmp BUN_INSTALL_CACHE_DIR=.bun-cache bun run build:deck
UV_CACHE_DIR=.uv-cache uv run scripts/export_slide_previews.py
```

Clean-Worktree Proof

At handoff, the intended proof standard is:

- the deck has been rebuilt
- previews have been regenerated
- the polish changes are committed
- `git status --short` is clean

Part 5: Model Audit (Internal)

> **Note:** These documents are for internal methodology review and transparency. They discuss alternative designs, sensitivity analyses, and historical model versions that are not part of the public-facing narrative.

Model Audit Documentation (INTERNAL)

Status: Internal documentation — not part of public release canon

This directory contains detailed audit materials for the forecasting model. These documents are:

- **For:** Internal verification, methodology review, and developer reference
- **Not for:** General public reading or citation as canonical project output

Contents

File	Purpose	Audience
`MODEL_AUDIT_COMPLETE.md`	Full technical audit of v1→v2 model changes	Technical reviewers
`MODEL_V2_DESIGN.md`	Design document for potential future enhancements	Model developers
`MODEL_V2_FORMAL_SPECIFICATION.md`	Mathematical formalization of V2 design	Technical reviewers
`ablation_validation_report.md`	Ablation study results	Methodology reviewers
`FLOOR_JUSTIFICATION_FRAMEWORK.md`	Rationale for floor parameters	Technical reviewers
`audit_*.png`	Diagnostic visualizations from audit process	Internal QA
`audit_data.json`	Machine-readable audit results	Internal tools

Public-Facing Model Description

For the canonical, public-facing explanation of the forecast model, see: - `../../content/06-forecast.md` — Main content explanation - `../../analyses/forecast_model.py` — Reference implementation - `../../data/forecast_scenarios.json` — Scenario parameters

Scope Note

The audit documents in this directory may discuss: - Alternative model designs not implemented - Sensitivity analyses and ablation studies - Known limitations and open questions - Historical model versions

These discussions are valuable for transparency but are **not** part of the project's public claims or findings. When citing this project, refer to the content in `content/` rather than these audit files.

This directory is part of the repository's internal quality assurance system.

Forecast Model Audit Package

Generated: 2026-03-25
Purpose: Provide complete, self-contained context for critical evaluation of the forecasting model
Scope: Both old (v1) and new (v2) model implementations

---

1. Executive Audit Summary

1.1 What the Old Model Did (Pre-March 2025)

The original model (v1) used simple multiplicative exponential growth on the observation scale:

```
y_t = y_{t-1} × exp(growth_mu + noise)
y_t = clip(y_t, 0, 100)  # hard bounds
```

Key characteristics: - Growth was exponential in raw scores (0-100) - Hard clipping at 0 and 100 created unnatural behavior at boundaries - Fixed growth rate per pillar (no evolution of growth over time) - Single noise source (no distinction between level and trend uncertainty)

Reported output: ~28% probability of crossing by 2040 (base case)

1.2 What the New Model Does (March 2025+)

The enhanced model (v2) uses a state-space approach with latent capability:

```
z_t = logit(y_t/100)                  # transform to unbounded scale
z_t = z_{t-1} + g_t + ε_t             # level evolves by growth rate plus noise
g_t = φ×g_{t-1} + (1-φ)×ḡ + η_t       # growth rate mean-reverts to long-run mean
y_t = 100 × sigmoid(z_t)              # transform back with natural saturation
```

Key characteristics: - Logit transform creates natural asymptotic approach to 100 (no hard clipping) - Growth rate g_t is stochastic and mean-reverting (φ = 0.9) - Two noise sources: level shock ε_t and growth shock η_t - Separates level uncertainty from trend uncertainty

Reported output: ~8-9% probability of crossing by 2040 (base case)

1.3 Key Structural Changes

Aspect	Old (v1)	New (v2)	Impact on Crossing
Scale	Raw 0-100	Logit → sigmoid	**MAJOR** - Natural saturation slows late-stage growth
Growth dynamics	Fixed exp	Local linear trend	**MINOR** - Mean reversion (φ=0.9) keeps growth near ḡ
Growth evolution	Constant	Mean-reverting	**NEGLIGIBLE** - φ sensitivity test shows almost no effect

Aspect	Old (v1)	New (v2)	Impact on Crossing
Noise structure	Single	Level + trend	**MODERATE** - Additional variance source

1.4 Attribution of the Difference

Primary driver of lower crossing probability: LOGIT/SIGMOID TRANSFORMATION

The ablation study reveals: - Model A (fixed growth, logit scale): 8.2% crossing - Model B (piecewise, 30% accel post-2028): 48.7% crossing - Model C (local linear trend, φ=0.9): 9.0% crossing

Critical finding: Model A and Model C produce nearly identical results (8.2% vs 9.0%). This means: 1. The mean-reverting growth (Model C's main feature) has MINIMAL impact on crossing probability 2. The logit/sigmoid transformation (shared by all models) is the DOMINANT factor 3. Model B's higher probability comes entirely from the 30% acceleration assumption, NOT from the state-space structure

Saturation analysis confirms this: - At score 20: logit gives 3.4 effective growth vs 2.0 linear (70% MORE growth) - At score 80: logit gives 3.0 effective growth vs 8.0 linear (62% LESS growth) - At score 90: logit gives 1.7 effective growth vs 9.0 linear (82% LESS growth) - At score 99: logit gives 0.2 effective growth vs 9.9 linear (98% LESS growth)

Conclusion: The model is more conservative primarily because high scores (70-100) grow MUCH slower under the sigmoid transformation, not because of sophisticated state-space dynamics.

1.5 What Has NOT Been Sufficiently Validated

1. **Whether logit is the appropriate transformation** - No theoretical justification provided beyond "natural saturation"
2. **Whether φ=0.9 is empirically grounded** - Sensitivity shows almost no effect, suggesting it may not matter
3. **Backtest against historical capability data** - No out-of-sample validation performed
4. **Whether floors or threshold is the binding constraint** - Floor sensitivity is high, threshold sensitivity is low
5. **Model B's 30% acceleration** - Arbitrary breakpoint year (2028) and acceleration magnitude
6. **Initial 2026 values** - Expert judgment, not fitted from data

---

2. Exact Model Documentation

2.1 Observation Scale

Pillar scores: Continuous values in [0, 100] - 0 = complete absence of capability - 100 = theoretical maximum / perfect capability - Initial values (2026) range from 20 (Network) to 55 (Capability)

2.2 Transformation Functions

Logit transform (observation → latent):

```
def to_logit(y):
    y_clipped = clip(y, 0.01, 99.99)  # prevent numerical issues
    return log(y / (100 - y))
```

Key values: | y | z = logit(y/100) | |—|————————| | 1 | -4.6 | | 20 | -1.39 | | 50 | 0 | | 80 | 1.39 | | 99 | 4.6 |

Sigmoid transform (latent → observation):

```
def from_logit(z):
    return 100 / (1 + exp(-z))
```

2.3 Latent State(s)

Model A (Fixed Log Growth): - Single latent state: z_t (capability level on logit scale)

Model B (Piecewise Growth): - Single latent state: z_t - Regime indicator: I(t < breakpoint) - Growth rate switches between g_pre and g_post

Model C (Local Linear Trend - DEFAULT): - Two latent states: - z_t: capability level (logit scale) - g_t: growth rate (change in z per year)

2.4 Evolution Equations

Model A - Fixed Log Growth:

```
 z_t = z_{t-1} + ḡ + ν_t
where ν_t ~ N(0, σ)
      ḡ = growth_mu (from config)
      σ = volatility (from config)
```

Model B - Piecewise Growth:

```
 z_t = z_{t-1} + g_t + ν_t
where g_t = g_pre if t < breakpoint else g_post
      g_pre = growth_mu
      g_post = growth_mu × 1.3 (30% acceleration)
      breakpoint = 2028 (hardcoded)
```

Model C - Local Linear Trend:

```
 Level equation:   z_t = z_{t-1} + g_t + ε_t
Growth equation:  g_t = φ×g_{t-1} + (1-φ)×ḡ + η_t

where ε_t ~ N(0, σ_level)    [level shock]
      η_t ~ N(0, σ_growth)   [growth shock]
      ḡ = growth_mu (from config)
      σ_level = volatility (from config)
      σ_growth = growth_volatility (scenario-dependent: 0.015-0.03)
      φ = 0.9 (persistence, hardcoded)
```

2.5 Observation Equation

All models:

```
 y_t = from_logit(z_t) = 100 / (1 + exp(-z_t))
```

2.6 Noise Components

Component	Symbol	Distribution	Source
Level shock	ε_t	N(0, σ_level)	volatility (config)
Growth shock	η_t	N(0, σ_growth)	growth_volatility (hardcoded per scenario)
Negative shock	-	Bernoulli(p) × size	neg_shock_prob, neg_shock_size (config)

Negative shock logic: Applied to level equation. If random < neg_shock_prob, add neg_shock_size (negative value) to growth.

2.7 Regime/Breakpoint Logic

Model A: No regime switching

Model B:

```
if year < 2028:
    growth = growth_mu
else:
    growth = growth_mu * 1.3  # 30% acceleration
```

Model C: No explicit regime, but mean reversion creates implicit regime toward long-run growth

2.8 Saturation/Begrenzing

Implicit via sigmoid: No explicit bounds checking needed. As z → ∞, y → 100 asymptotically.

Numerical clipping: In to_logit(), y is clipped to [0.01, 99.99] to prevent log(0) or division by zero.

2.9 Crossing Logic

Threshold: Readiness Index ≥ 75

Index calculation (all models):

```
Index_t = 100 × Π_i (pillar_i,t / 100)^weight_i
```

Weights: - C (Capability): 20% - E (Efficiency): 20% - M (Memory): 15% - R (Reliability): 15% - N (Network): 12% - G (Governance): 10% - D (Demand): 8%

Floors: Four pillars must each be ≥ 60: - M (Memory) ≥ 60 - R (Reliability) ≥ 60 - N (Network) ≥ 60 - G (Governance) ≥ 60

Consecutive years rule: Threshold AND floors must be met for 2 consecutive years before crossing is declared.

Algorithm:

```
crossing_year = NaN
for each simulation:
    consecutive = 0
    for year_idx, year in enumerate(years):
        floors_met = all(pillar_values[code][sim][year_idx] >= floor for code, floor in floors)
        if index[sim][year_idx] >= threshold and floors_met:
            consecutive += 1
            if consecutive >= 2:
                crossing_year = year
                break
        else:
            consecutive = 0
```

---

3. File Inventory

3.1 Core Model Files

File	Path	Purpose	Key Assumptions
Main model	`analyses/forecast_model.py`	Complete implementation of A, B, C models	φ=0.9, breakpoint=2028, growth_volatility scenario-dependent
Extended model	`analyses/forecast_model_v2.py`	Comparison and visualization	Same as main
Config	`data/forecast_scenarios.json`	All parameters	Initial values, growth_mu, volatility, floors, threshold are EXPERT JUDGMENT, not fitted

3.2 Documentation Files

File	Path	Purpose
Content	`content/06-forecast.md`	Explains model to readers
AGENTS.md	`AGENTS.md`	Project overview
This audit	`docs/model_audit/MODEL_AUDIT_COMPLETE.md`	Truth-first audit
Audit data	`docs/model_audit/audit_data.json`	Machine-readable ablation results

3.3 Generated Assets

File	Path	Purpose
Main plot	`assets/forecast_distribution.png`	4-panel visualization
Comparison	`assets/forecast_model_comparison_*.png`	Model A/B/C comparison
Uncertainty	`assets/forecast_uncertainty_decomposition.png`	Level vs trend uncertainty

File	Path	Purpose
Sigmoid demo	`assets/forecast_sigmoid_demo.png`	Mathematical transformation

---

4. Code Excerpts

4.1 Transformation Functions

```
def to_logit(y: np.ndarray) -> np.ndarray:
    """Transform 0-100 scores to logit scale.

    Mathematical definition:
        z = log(y / (100 - y))

    This maps:
        y=50  → z=0
        y=90  → z≈2.2
        y=99  → z≈4.6
        y→100 → z→∞ (asymptotic saturation)
    """
    y_clipped = np.clip(y, 0.01, 99.99)  # GUARDRAIL: prevent numerical issues
    return np.log(y_clipped / (100 - y_clipped))


def from_logit(z: np.ndarray) -> np.ndarray:
    """Transform logit scale back to 0-100 scores.

    Mathematical definition:
        y = 100 / (1 + exp(-z)) = 100 * sigmoid(z)
    """
    return 100.0 / (1.0 + np.exp(-z))
```

Location: lines 46-68 in `analyses/forecast_model.py`

4.2 Model C - Local Linear Trend (Default)

```
class ModelC_LocalLinearTrend(GrowthModel):
    """Model C: Local linear trend state-space model (DEFAULT).

    State equations on logit scale:
        z_t = z_{t-1} + g_t + eps_t      (level equation)
        g_t = phi*g_{t-1} + (1-phi)*g_bar + eta_t  (growth equation)

    Where:
        z_t = latent capability level
        g_t = latent growth rate (mean-reverting to g_bar)
        phi = persistence parameter (default 0.9)
        eps_t ~ N(0, sigma_level), eta_t ~ N(0, sigma_growth)
    """

    def __init__(self, rng: np.random.Generator, persistence: float = 0.9):
        super().__init__(rng)
        self.persistence = persistence  # phi = 0.9 HARDCODED
```

```python
    def simulate_pillar(self, pillar: PillarConfig, n_simulations: int, n_years: int) -> np.ndarray:
        """Simulate with local linear trend on logit scale."""
        z_values = np.zeros((n_simulations, n_years))
        z_values[:, 0] = to_logit(pillar.initial_2026)

        # Initialize latent growth rate with uncertainty
        g_values = np.full(n_simulations, pillar.growth_mu) + self.rng.normal(
            0, 0.05, n_simulations  # INITIAL UNCERTAINTY: 0.05 std
        )

        sigma_level = pillar.volatility      # From config
        sigma_growth = pillar.growth_volatility  # HARDCODED per scenario

        for t in range(1, n_years):
            # Growth equation with mean reversion
            g_values = (
                self.persistence * g_values
                + (1 - self.persistence) * pillar.growth_mu
                + self.rng.normal(0, sigma_growth, n_simulations)
            )

            # Level equation
            level_shock = self.rng.normal(0, sigma_level, n_simulations)
            shocks = self.rng.random(n_simulations) < pillar.neg_shock_prob
            level_shock[shocks] += pillar.neg_shock_size

            z_values[:, t] = z_values[:, t - 1] + g_values + level_shock

        return from_logit(z_values)
```

Location: lines 235-286 in `analyses/forecast_model.py`

4.3 Threshold Crossing Logic

```python
def _check_threshold_crossing(
    self,
    index: np.ndarray,           # Shape: (n_simulations, n_years)
    pillar_values: dict[str, np.ndarray],
    threshold: float | None = None,
    floor_values: dict[str, float] | None = None,
) -> np.ndarray:
    """Determine the first year a simulation crosses and sustains the threshold."""
    active_threshold = self.scenario.threshold if threshold is None else threshold
    active_floors = self.scenario.floor_values if floor_values is None else floor_values
    crossing_years = np.full(self.n_simulations, np.nan)

    for sim_idx in range(self.n_simulations):
        consecutive = 0
        for year_idx, year in enumerate(self.years):
            # Check ALL floors
            floors_met = all(
                pillar_values[code][sim_idx, year_idx] >= minimum
                for code, minimum in active_floors.items()
            )
            # Check threshold AND floors
```

```
            if index[sim_idx, year_idx] >= active_threshold and floors_met:
                consecutive += 1
                if consecutive >= self.scenario.min_consecutive_years:  # 2 years
                    crossing_years[sim_idx] = year
                    break
            else:
                consecutive = 0  # RESET if either condition fails

    return crossing_years
```

Location: lines 328-355 in `analyses/forecast_model.py`

4.4 Index Calculation

```
def _calculate_readiness_index(self, pillar_values: dict[str, np.ndarray]) -> np.ndarray:
    """Calculate the weighted geometric mean readiness index."""
    index = np.ones((self.n_simulations, len(self.years)))
    for pillar in self.scenario.pillars:
        index *= (pillar_values[pillar.code] / 100.0) ** pillar.weight
    return index * 100.0
```

Key property: Uses geometric mean, not arithmetic. Low pillars drag down the index more than high pillars pull it up.

Location: lines 321-326 in `analyses/forecast_model.py`

---

5. Historical Data Context

5.1 What Historical Data Is Used

Answer: NONE for the forecasting itself.

The model is purely assumption-driven. The parameters are: - Initial 2026 values: Expert judgment - Growth rates (growth_mu): Expert judgment - Volatilities: Expert judgment - Floors and threshold: Expert judgment

5.2 What External Data Informs the Model

The following sources inform the parameter choices but are NOT used to fit the model:

Source	Data	How It Informs
Stanford HAI AI Index 2025	Benchmark improvements (MMMU +18.8, GPQA +48.9, SWE-bench +67.3)	Suggests capability growth is possible
Epoch AI	Compute growth (~5x/year), efficiency (~3x/year)	Informs Efficiency pillar assumptions

Source	Data	How It Informs
METR	Time horizon doubling ~7 months	Suggests accelerating capability possible

Critical caveat: These data points are used to SET PARAMETERS, not to VALIDATE the model. There is no backtest against historical pillar trajectories.

5.3 Data Sample

N/A - No time series data is used. The model is initialized at 2026 with point estimates.

5.4 Which Assumptions Are Expert Judgment vs Data-Derived

Parameter	Source	Confidence
Initial 2026 values (C=55, E=45, etc.)	Expert judgment	LOW - Arbitrary starting points
Growth rates (growth_mu)	Expert judgment + trend data	MEDIUM - Informed by AI Index but not fitted
Volatility	Expert judgment	LOW - No empirical basis
Floors (60)	Expert judgment	MEDIUM - Based on institutional analysis
Threshold (75)	Expert judgment	MEDIUM - Based on qualitative readiness
φ (0.9)	Hardcoded	LOW - Not fitted or validated
Breakpoint (2028)	Hardcoded	LOW - Arbitrary
Acceleration (30%)	Hardcoded	LOW - Arbitrary

---

6. Parameter Provenance

6.1 Complete Parameter Table

Parameter	Current Value	Defined In	Meaning	Source	Sensitivity
Weights					
C_weight	0.20	forecast_scenarios.json	Capability importance	Expert judgment	LOW (fixed)
E_weight	0.20	forecast_scenarios.json	Efficiency importance	Expert judgment	LOW (fixed)
M_weight	0.15	forecast_scenarios.json	Memory importance	Expert judgment	LOW (fixed)
R_weight	0.15	forecast_scenarios.json	Reliability importance	Expert judgment	LOW (fixed)

Parameter	Current Value	Defined In	Meaning	Source	Sensitivity
N_weight	0.12	forecast_scenarios.json	Network importance	Expert judgment	LOW (fixed)
G_weight	0.10	forecast_scenarios.json	Governance importance	Expert judgment	LOW (fixed)
D_weight	0.08	forecast_scenarios.json	Demand importance	Expert judgment	LOW (fixed)
Initial Values (2026)					
C_initial	55	forecast_scenarios.json	Starting capability	Expert judgment	HIGH
E_initial	45	forecast_scenarios.json	Starting efficiency	Expert judgment	HIGH
M_initial	30	forecast_scenarios.json	Starting memory	Expert judgment	HIGH
R_initial	28	forecast_scenarios.json	Starting reliability	Expert judgment	HIGH
N_initial	20	forecast_scenarios.json	Starting network	Expert judgment	HIGH
G_initial	25	forecast_scenarios.json	Starting governance	Expert judgment	HIGH
D_initial	50	forecast_scenarios.json	Starting demand	Expert judgment	HIGH
Growth Rates (ḡ)					
C_growth	0.18	forecast_scenarios.json	Capability growth	Expert + trend data	HIGH
E_growth	0.28	forecast_scenarios.json	Efficiency growth	Expert + trend data	HIGH
M_growth	0.14	forecast_scenarios.json	Memory growth	Expert judgment	HIGH
R_growth	0.16	forecast_scenarios.json	Reliability growth	Expert judgment	HIGH
N_growth	0.15	forecast_scenarios.json	Network growth	Expert judgment	HIGH
G_growth	0.12	forecast_scenarios.json	Governance growth	Expert judgment	HIGH
D_growth	0.12	forecast_scenarios.json	Demand growth	Expert judgment	HIGH

Parameter	Current Value	Defined In	Meaning	Source	Sensitivity
Noise Parameters					
C_volatility	0.08	forecast_scenarios.json	Capability noise	Expert judgment	MEDIUM
σ_growth (base)	0.02	forecast_model.py	Growth rate volatility	HARDCODED	LOW (tested)
φ (persistence)	0.9	forecast_model.py	Mean reversion strength	HARDCODED	VERY LOW
Threshold/Floors					
Threshold	75	forecast_scenarios.json	Readiness index target	Expert judgment	LOW
Floor M,R,N,G	60	forecast_scenarios.json	Minimum pillar scores	Expert judgment	HIGH
Consecutive years	2	forecast_scenarios.json	Sustained crossing rule	Expert judgment	MEDIUM
Model B Specific					
Breakpoint year	2028	forecast_model.py	Regime switch	HARDCODED	NOT TESTED
Post-breakpoint accel	30%	forecast_model.py	Growth acceleration	HARDCODED	NOT TESTED

6.2 Key Uncertainties

HIGH SENSITIVITY (changes crossing probability by >5%): - Initial values (especially low ones like N=20, G=25) - Growth rates (especially slow ones like G=0.12) - Floor values (60 is critical; 55→65 changes crossing from 17% to 3%)

MEDIUM SENSITIVITY (changes crossing probability by 1-5%): - Volatility parameters - Threshold (only at extreme values like 85)

LOW/NEGLIGIBLE SENSITIVITY (changes crossing probability by <1%): - φ (persistence) - surprisingly invariant - σ_growth (growth volatility) - Weights (fixed, not tested)

---

7. Crossing Attribution (Ablation Study)

7.1 Full Ablation Results

Base case scenario, 10,000 simulations:

Model	P(cross by 2040)	Median year (if crosses)	Never crosses
A (Fixed log growth)	8.2%	2040	91.8%
B (Piecewise, +30% post-2028)	48.7%	2039	51.3%
C (Local linear trend, φ=0.9)	9.0%	2039	91.0%

7.2 Step-by-Step Attribution

Starting from Model A (8.2%), what changes increase/decrease crossing probability?

Change	Effect	Explanation
Logit transform (vs raw exponential)	-20% (estimated)	Saturates high scores, slowing late-stage growth
Mean reversion (φ=0.9)	+0.8%	Negligible - growth stays near ḡ
Separate growth noise (σ_growth)	+0-1%	Adds variance, minimal net effect
Combined (Model C)	**+0.8%**	Effectively same as Model A
30% acceleration post-2028	**+40.5%**	Massive effect if acceleration assumed

7.3 Critical Insight

The “enhanced” state-space model (Model C) produces virtually identical results to the “simple” fixed-growth model (Model A).

This suggests: 1. The sophisticated state-space structure may be unnecessary complexity 2. The logit/sigmoid transformation is doing ALL the work in making the model conservative 3. The mean reversion (φ=0.9) is so strong that growth rates don’t meaningfully evolve

7.4 Floor vs Threshold Sensitivity

Floor value	P(cross by 2040)	Change from baseline
50	26.6%	+18.4pp
55	17.1%	+8.9pp
60 (baseline)	8.2%	-
65	3.1%	-5.1pp
70	1.0%	-7.2pp

Threshold	P(cross by 2040)	Change from baseline
70	8.3%	+0.1pp
75 (baseline)	8.2%	-

Threshold	P(cross by 2040)	Change from baseline
80	5.8%	-2.4pp
85	0.6%	-7.6pp

Finding: Floors are the binding constraint at current settings. The threshold only becomes limiting at extreme values (85+).

---

8. Debug / Sanity Checks

8.1 Scale Consistency After Logit/Sigmoid

Test: Transform [1, 20, 50, 80, 99] to logit and back.

Original y	z = logit(y)	y' = sigmoid(z)	Error
1	-4.595	1.0	0%
20	-1.386	20.0	0%
50	0.0	50.0	0%
80	1.386	80.0	0%
99	4.595	99.0	0%

Result: ✓ PASS - Transformations are mathematically inverse

8.2 Saturation Timing

Question: Does saturation kick in too early?

Evidence from saturation analysis: - At y=70: logit effective growth = 4.0 vs linear 7.0 (43% reduction) - At y=80: logit effective growth = 3.0 vs linear 8.0 (62% reduction) - At y=90: logit effective growth = 1.7 vs linear 9.0 (82% reduction)

Assessment: Growth slows significantly above 70-80. This is the dominant reason for low crossing probability.

8.3 Growth Mean Reversion Strength

Test: Vary φ from 0.7 to 0.99

φ	P(cross by 2040)
0.7	8.5%
0.8	8.4%
0.9	8.2%
0.95	8.3%
0.99	8.1%

Result: ALMOST NO EFFECT. This is suspicious.

Hypothesis: The level noise (σ_level) and growth noise (σ_growth) are calibrated such that g_t doesn't deviate far from ḡ over the 15-year horizon. Mean reversion doesn't matter if growth never has time to drift.

8.4 Floors Correctly Applied

Test: Check that floor constraint requires ALL four pillars (M,R,N,G) ≥ floor.

Code verification:

```
floors_met = all(
    pillar_values[code][sim_idx, year_idx] >= minimum
    for code, minimum in active_floors.items()
)
```

Result: ✓ PASS - Uses `all()`, not `any()` or average

8.5 Consecutive Years Logic

Test: Check that 2 consecutive years are required and that counter resets on failure.

Code verification:

```
if index >= threshold and floors_met:
    consecutive += 1
    if consecutive >= 2:  # CHECK: >= not ==
        crossing_year = year
        break
else:
    consecutive = 0  # CHECK: resets to 0
```

Result: ✓ PASS - Logic is correct

8.6 Initial State

Test: Verify 2026 values match config.

Code verification:

```
z_values[:, 0] = to_logit(pillar.initial_2026)
```

Result: ✓ PASS - Uses config values directly

8.7 Off-by-One in Years

Test: Check that simulation runs 2026-2040 inclusive (15 years).

Code:

```
self.years = np.arange(2026, 2041)  # 15 years: 2026, ..., 2040
```

Result: ✓ PASS - 15 years total

8.8 Unexpected Clipping

Test: Check for hidden clip operations.

Findings: - `to_logit()` : clips y to [0.01, 99.99] to prevent log(0) - `from_logit()` : No clipping needed (sigmoid naturally bounded) - Main simulation: No explicit clipping

Result: ✓ PASS - Only numerical guardrails

8.9 Percentile Calculations

Test: Verify median, p5, p95 use correct axis.

Code:

```
median = np.median(index, axis=0)  # Across simulations
p25 = np.percentile(index, 25, axis=0)
```

Result: ✓ PASS - axis=0 means across simulations, per year

8.10 Negative Shock Application

Test: Verify negative shocks are applied to level, not growth.

Code:

```
level_shock = self.rng.normal(0, sigma_level, n_simulations)
shocks = self.rng.random(n_simulations) < pillar.neg_shock_prob
level_shock[shocks] += pillar.neg_shock_size  # Added to LEVEL shock
z_values[:, t] = z_values[:, t - 1] + g_values + level_shock
```

Result: ✓ PASS - Applied to level equation

---

9. Backtests and Model Comparison

9.1 The Fundamental Problem

There is no historical data on the 7 pillars to backtest against.

The pillars are conceptual constructs, not measured time series. We cannot: - Test out-of-sample forecasts - Compute RMSE against historical trajectories - Validate the logit transform choice empirically

9.2 What Could Be Done (But Wasn't)

Test	Data Needed	Status
Rolling origin on pillar trajectories	Time series of C,E,M,R,N,G,D scores	NOT AVAILABLE
Proxy validation using ECI	Epoch Capability Index 2023-2025	NOT DONE

Test	Data Needed	Status
Compute-growth correlation	Training compute vs capability	NOT DONE
Expert elicitation validation	Survey of AI researchers	NOT DONE

9.3 Qualitative Model Comparison

Criterion	Model A	Model B	Model C
Mathematical simplicity	✓✓✓	✓✓	✓
Natural saturation	✓	✓	✓
Growth rate evolution	×	× (step)	✓ (smooth)
Mean reversion	×	×	✓
Computational cost	Low	Low	Medium
Crossing probability (base)	8.2%	48.7%	9.0%

9.4 Which Model is “Best”?

From a forecasting perspective: Cannot determine without validation data.

From a theoretical perspective: - Model C has the most plausible microfoundations (stochastic growth, mean reversion) - Model B’s 30% acceleration is arbitrary - Model A is too simple but produces similar results to Model C

From a practical perspective: - Model A and C are essentially equivalent in output (8.2% vs 9.0%) - Model C adds complexity without changing conclusions - Model B’s high probability (48.7%) depends entirely on the arbitrary 30% acceleration

---

10. Visualizations

10.1 Generated Plots

File	Description	Key Insight
`forecast_distribution.png`	4-panel main visualization	Shows trajectories, crossing CDF, distribution, audit notes
`forecast_model_comparison_*.png`	Model A/B/C comparison per scenario	Model B dramatically different due to acceleration
`forecast_uncertainty_decomposition.png`	Level vs trend uncertainty	Most uncertainty is level, not trend
`forecast_sigmoid_demo.png`	Mathematical transformation	Visual proof of early acceleration, late saturation

10.2 Recommended Additional Plots

1. **Pillar trajectory fan charts** - Show each pillar's distribution over time
2. **Floor binding analysis** - Which floor fails most often
3. **Sensitivity tornado diagram** - Ordered parameter sensitivities
4. **Historical context plot** - If any proxy data becomes available

---

11. Open Questions / Risks

11.1 What Is Still Uncertain

1. **Appropriateness of logit transform** - Why logit and not probit, log-normal, or something else?
2. **Initial 2026 values** - Are N=20, G=25 realistic? No data to validate.
3. **Growth rate calibration** - Why is G (Governance) only 0.12 while E (Efficiency) is 0.28?
4. **φ=0.9 justification** - If mean reversion has no effect on results, why include it?
5. **Model B's breakpoint** - Why 2028? Why 30% acceleration?

11.2 Which Choices Are Most Discutable

Rank	Choice	Issue	Severity
1	Logit saturation at high scores	80→90 is extremely slow; is this realistic?	HIGH
2	Floor at 60	Sensitivity is extreme; small changes → large output changes	HIGH
3	Initial N=20, G=25	Very low starting points; is 2026 realistic?	MEDIUM
4	Model B's 30% acceleration	Arbitrary magnitude and timing	MEDIUM
5	φ=0.9	No empirical basis; sensitivity shows it doesn't matter	LOW

11.3 What Is Likely Correct

1. **Geometric mean for index** - Low pillars should drag down overall readiness
2. **Floors on M,R,N,G** - Institutional factors are necessary, not just nice-to-have
3. **Consecutive years rule** - Prevents one-off lucky spikes
4. **Monte Carlo approach** - Uncertainty quantification is essential
5. **Separating level and trend noise** - Conceptually correct even if empirically unvalidated

11.4 Highest Risk of Model Misspecification

PRIMARY RISK: The logit/sigmoid transformation may be TOO conservative at high scores.

Evidence: - Effective growth at y=90 is 82% lower than linear - Effective growth at y=95 is 91% lower than linear - This creates an "asymptotic trap" where reaching 100 becomes nearly impossible

SECONDARY RISK: The floors may be too high relative to initial values.

Evidence: - N starts at 20, must reach 60 (40-point gap) - G starts at 25, must reach 60 (35-point gap) - With logit saturation, this requires enormous latent growth

11.5 Tests That Are Missing

1. **Historical proxy validation** - Can we map any existing metrics to pillars?
2. **Expert elicitation** - Do AI researchers agree with initial values and growth rates?
3. **Alternative transforms** - How would probit, log-normal, or power transforms change results?
4. **Alternative saturation functions** - What if saturation kicks in at 90 instead of 80?
5. **Stress testing** - What parameter combinations produce crossing by 2030? Are any plausible?

---

12. Questions ChatGPT Should Challenge

12.1 On Model Structure

1. **Is the sigmoid saturation at high scores (80-100) empirically justified or mathematically convenient?**
2. **Why use logit/sigmoid rather than a learned saturation function or a power transform?**
3. **If φ (persistence) sensitivity is negligible (0.7→0.99 changes crossing by <1%), why include mean reversion at all?**
4. **Does the state-space structure (Model C) add value if it produces nearly identical results to fixed growth (Model A)?**
5. **Is the geometric mean index appropriate, or should low pillars "fail" the system rather than just drag it down?**

12.2 On Parameters

6. **Are the initial 2026 values (especially N=20, G=25) too low to be realistic?**
7. **Why is Governance growth (0.12) less than half of Efficiency growth (0.28)? Is this pessimism justified?**
8. **Is the floor at 60 too high given initial values? Would 55 or 50 be more realistic?**
9. **What empirical evidence supports φ=0.9 specifically?**
10. **For Model B, what justifies 2028 as the breakpoint and 30% as the acceleration magnitude?**

12.3 On Validation

11. **Is there any historical time series that could proxy for even one pillar to enable backtesting?**
12. **How would the model change if we used ECI (Epoch Capability Index) as a validation target?**
13. **Has expert elicitation been done to validate the initial values and growth assumptions?**
14. **What would a "stress test" scenario look like that achieves 50% crossing by 2035, and is it plausible?**

12.4 On Interpretation

15. **If Model A (simple) and Model C (complex) agree at 8-9% crossing, what confidence should we have in this number?**
16. **Does the 91% “never crosses” rate mean the model is pessimistic or that true readiness is genuinely hard?**
17. **How should decision-makers use a forecast that says “probably not by 2040”?**

Appendix A: Audit Data (Machine Readable)

File: `docs/model_audit/audit_data.json`

Contains: - `ablation_study`: Crossing probabilities for Models A, B, C - `parameter_sensitivity`: Crossing probabilities for varying φ, threshold, floors - `saturation_analysis`: Effective growth rates at different score levels

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Appendix B: File Checksums (for Reproducibility)

File	SHA256 (first 16 chars)
forecast_model.py	(generate if needed)
forecast_scenarios.json	(generate if needed)

END OF AUDIT PACKAGE

Truth status: This document aims for complete honesty about what is known, assumed, and uncertain. All parameter sensitivities and ablation results are generated from actual code execution, not hand-waved.

Last updated: 2026-03-25

Floor Justification Framework

Status: Research-backed recommendations for making floors=60 defensible
Based on: Maturity frameworks (TRL, CMMI), composite indicator methodology, structured expert elicitation

---

De Kernvraag

> "Waarom 60?"

De ablatie toont dat dit de dominante vraag is. Deze document biedt een framework om die vraag te beantwoorden zonder het model te forceren.

Twee Coherente Interpretaties van Floors

Optie 1: Floors als Definitional "Stage Gate"

Visie: Floors zijn geen empirische parameters - ze definieren wat we bedoelen met "crossing".

Analogie: NASA Technology Readiness Levels (TRL) - TRL 1-3: Basic principles observed - TRL 4-5: Validation in laboratory/relevant environment
- TRL 6-7: System/subsystem demonstration - TRL 8-9: System complete, mission proven

Toepassing op onze pillars:

Pillar	TRL-equivalent bij 60	Wat het betekent
G (Governance)	TRL 5-6	Governance framework geïmplementeerd, getest in relevante omgeving
N (Network)	TRL 5-6	Coördinatieprotocol stabiel in pilot deployments
M (Memory)	TRL 5-6	State persistence betrouwbaar in productie-achtige settings
R (Reliability)	TRL 5-6	Betrouwbaarheid gedemonstreerd over langere periodes

Voordelen: - Maakt "60" interpreteerbaar en debatteerbaar - Sluit aan bij bestaande engineering practices - Niet geclaimd als "waarheid" maar als "specificatie"

Nadeel: - Vereist nog steeds argumentatie waarom TRL 5-6 het juiste niveau is

Optie 2: Floors als Empirische Parameters (met Uncertainty)

Visie: Floors schatten het minimale niveau voor stabiele multi-agent deployment, maar we weten dit onzeker.

Methode: Structured Expert Elicitation - Geen puntinschattingen - Distributies per pillar (bijv. 50e, 25e, 75e percentiel) - Monte Carlo samplet uit deze distributies - Resultaat: confidence intervals ipv puntinschattingen

Praktisch protocol: 1. Definieer “floor meaning” per pillar als observable claims - G: “auditable risk management, lifecycle controls, accountability institutionalized” - N: “protocol stability under realistic load, fault tolerance demonstrated” - M: “state reconstruction reliability >X% over Y-day periods” - R: “error rates <Z% in production-like conditions”

2. Eliciteer distributies van 5-10 experts
    - “Wat is het minimale governance-niveau voor stabiele agentsamenleving?”
    - Geef median + 90% CI, niet punt
3. Run model met gesamplede floors
    - Resultaat: range van crossing probabilities
    - Bijv. “9% by 2040 (als floors=60), maar 17% (als floors=55)”

Voordelen: - Houdt onzekerheid expliciet - Kan bijgesteld worden naarmate meer data beschikbaar komt - Eerlijk over epistemische limieten

---

Maturity Frameworks voor Anchoring

NASA Technology Readiness Levels (TRL)

TRL	Beschrijving	Onze schaal (geschat)
1	Basic principles observed	0-20
2-3	Technology concept formulated/analyzed	20-40
4	Validation in laboratory	40-50
5	**Validation in relevant environment**	**50-65**
6	**System/subsystem demonstration**	**60-75**
7	System prototype in operational environment	75-85
8-9	System complete, mission proven	85-100

Claim: Floor=60 komt overeen met “TRL 5-6 grens”: - Voorbij lab proof-of-concept - In relevante omgeving gedemonstreerd - Maar nog niet volledig operationeel bewezen

CMMI Staged Levels (voor G - Governance)

CMMI Level	Beschrijving	Governance Floor
1	Initial (chaotic)	< 40
2	Managed (project-level)	40-50
3	**Defined (organization-level)**	**50-65**
4	Quantitatively managed	65-80
5	Optimizing	80-100

Claim: Governance=60 komt overeen met “Defined”: - Processen gestandaardiseerd op organisatie-niveau - Proactief management, niet alleen reactief - Maar nog niet volledig data-gedreven optimalisatie

ISO/IEC 42001 & NIST AI RMF (voor G)

ISO/IEC 42001: AI Management System (AIMS) requirements - Governance als cross-cutting function - Risk management geïntegreerd in lifecycle - Continuous improvement

NIST AI RMF: - GOVERN: governance als hoofdfunctie - MAP/MEASURE/MANAGE: ondersteunende functies

Operationalisatie naar score 60: - G=60: “AIMS requirements gedeeltelijk geïmplementeerd, governance framework actief maar nog niet volledig mature”

Horizon Effecten als Right-Censoring

Het Probleem

“91% never crosses by 2040” is misleidend. Het is **right-censored**: we weten alleen dat het event NA 2040 plaatsvindt, niet dat het nooit plaatsvindt.

Oplossing: Survival Analysis Reporting

Huidig (misleading):

```
P(cross by 2040) = 9%
"91% never crosses"
```

Gewenst (honest):

```
Jaar    P(cross by year)    P(nog niet gecrossed)
2030    0.1%                99.9%
2035    2.3%                97.7%
2040    9.0%                91.0%  ← censored
2045    37.5%               62.5%
2050    62.4%               37.6%
2060    88.1%               11.9%

"Median crossing time: 2047 (unconditional)"
"Of: 2039 (conditional on crossing by 2060)"
```

Visualisatie: Survival curve (Kaplan-Meier stijl) ipv enkele percentages

---

Concrete Actiepunten

Onmiddellijk (zonder model aanpassing)

1. **Vervang “never crosses” taal**
   - In content, slides, rapportage
   - Altijd horizon vermelden: “not by 2040”, niet “never”
2. **Presenteer multi-horizon cumulatieve distributie**
   - 2040, 2045, 2050, 2060
   - Maakt right-censoring expliciet
3. **Documenteer floors als specifieke keuze**

    - Niet “het model zegt”
    - Maar: “gegeven floors=60 als definitie van readiness…”

Korte termijn (met content updates)

4. **Voeg TRL/CMMI mapping toe aan documentatie**
    - Leg uit: 60 = TRL 5-6 = “relevant environment demonstration”
    - Maakt de keuze interpreteerbaar
5. **Presenteer floor sensitivity als scenario**
    - Strikte definitie (floors=65): X% by 2040
    - Moderate definitie (floors=60): 9% by 2040
    - Loose definitie (floors=55): 17% by 2040
    - Laat lezer kiezen welke definitie passend is

Lange termijn (indicatief)

6. **Structured expert elicitation**
    - Voor floors als empirische parameters
    - Distributies ipv puntwaarden
    - Vermeidt false precision
7. **Proxy time series voor pillars**
    - Oxford AI Readiness Index → G proxy
    - OECD AI indicators → governance/policy proxy
    - Update floor beliefs naarmate data beschikbaar komt

Research-Backed Referenties

Bron	Relevantie
NASA TRL Handbook	Maturity framework voor floors als definitional stage gates
CMMI Model	Governance maturity levels (G pillar)
ISO/IEC 42001	AI management system requirements (operationalisatie G)
NIST AI RMF	Governance framework specificatie
Cooke (1991)	Structured expert judgment voor uncertainty quantification
OECD/JRC Handbook	Composite indicator methodology (validatie van onze ablatie aanpak)
Kaplan-Meier (1958)	Survival analysis voor right-censoring

Samenvatting

De vraag “waarom 60?” heeft twee valide antwoorden:

1. **Definitional:** “60 = TRL 5-6 = het punt waarop we iets ‘echt’ noemen ipv ‘proof of concept’”
    - Dan is het een spec, geen voorspelling
    - En we tonen sensitivity: wat gebeurt er bij 55 of 65?
2. **Empirisch:** “We schatten dat 60 het minimale niveau is, maar zijn onzeker”

- Dan moeten we die onzekerheid propageren
- En resultaat is een range, geen punt

Wat we NU doen: - Eerlijk zeggen: floors=60 is een keuze die de uitkomst domineert - Multi-horizon rapporteren (geen “never”) - TRL/CMMI mapping toevoegen om 60 interpreteerbaar te maken

Wat we LATER kunnen doen: - Expert elicitation voor uncertainty - Proxy data voor empirische update

---

Status: Framework complete. Ready for content implementation.

Ablation Validation Report

Generated: 2026-03-25
Purpose: Truth-first validation of ablation results, correcting errors and overstated claims

---

Corrections to Previous Summary

Error 1: Incorrect 2-of-3 claim

What I said: “2-of-3 jaren ipv 2 strikt → 14.7% crossing”

What the data actually shows: - 1 jaar: **14.7%** - 2 jaar (strict): **9.0%** - 2-of-3: **9.2%**

Correction: The 2-of-3 rule produces **9.2%**, not 14.7%. It is essentially identical to the strict 2-year rule (difference: 0.2pp). This suggests that most failures to cross are not “just missed by 1 year” but more fundamental misses.

Error 2: “80% of conservatism” claim was too strong

What I said: “Floors are responsible for ~80% of the conservatism”

What the data actually supports: - No floors: 47.1% crossing - Baseline (floors 60): 9.0% crossing - Difference: **38.1 percentage points**

Correction: Floors reduce crossing probability by **38pp**, which is the largest single effect, but “80% of conservatism” is not a well-defined claim. Better formulation:

> “Within this parameterization, floors are the dominant structural constraint, reducing crossing probability by 38pp compared to an index-only specification.”

Error 3: “Shocks have 0 impact on median” was poorly formulated

What I said: “Shocks have 0 impact on median”

What the data actually shows: - Deterministic crossing: **2039** - Stochastic median (among successes): **2039**

Correction: Shocks affect the **distribution shape** and **success rate**, but do not shift the **central tendency of successful crossings**. This is different from “no impact”—it means the median successful run is not systematically delayed or accelerated by noise, but many runs fail entirely due to adverse shock sequences.

Corrected Ablation Results

1. Floor Architecture (validated)

Configuration	P(cross by 2040)	Change from baseline
No floors	47.1%	+38.1pp

Configuration	P(cross by 2040)	Change from baseline
Floors 50	27.1%	+18.1pp
Floors 55	17.4%	+8.4pp
Floors 60 (baseline)	**9.0%**	—
Floors 65	3.6%	-5.4pp

Finding: Floors are the dominant constraint. The effect is monotonic and large.

2. Time Horizon (validated)

Horizon	P(cross by horizon)	P(cross by 2040)	Median year (if crosses)
2040	9.0%	9.0%	2039
2045	37.5%	9.3%	2042
2050	62.4%	8.8%	2045
2060	88.1%	8.4%	2047

Finding: The “91% never crosses by 2040” is misleading. By 2050, 62% have crossed. The horizon truncation creates an artificial “never” category.

3. Consecutive Years Rule (corrected)

Rule	P(cross by 2040)	Change from baseline
1 year	14.7%	+5.7pp
2 years strict (baseline)	**9.0%**	—
2-of-3	9.2%	+0.2pp
Rolling 2-year	9.0%	0pp

Finding: The 2-year rule reduces crossing by 5.7pp vs 1-year. The 2-of-3 variant provides almost no relaxation (0.2pp), suggesting that failures are not marginal “near misses” but substantial misses.

4. Aggregation Method (validated)

Method	P(cross by 2040)	Change from baseline
Geometric (baseline)	**9.0%**	—
Arithmetic	9.1%	+0.1pp
CES (rho=-1)	8.9%	-0.1pp
Softmin	5.6%	-3.4pp

Finding: Geometric vs arithmetic makes negligible difference (0.1pp). Only softmin (which heavily penalizes low values) has a substantial effect. The “double penalty” hypothesis (geometric mean + floors) is partially incorrect

—floors completely dominate.

5. Deterministic Baseline (validated)

Deterministic trajectory (median growth, no shocks): - All pillars reach 60 by 2039 - Index reaches 75+ by 2039 - **Deterministic crossing: 2039**

Finding: The deterministic path crosses at the same year as the stochastic median (2039). Noise affects the probability of success, not the central timing of success.

Honest Interpretation

What this actually shows

1. **The model is essentially a floor-gated model**
    - Threshold 75 is rarely binding (floors fail first)
    - Floor architecture explains ~38pp of the 9% baseline
    - Aggregation method is almost irrelevant
2. **The 2040 horizon creates artificial truncation**
    - “Never crosses” should be “not by 2040”
    - By 2050, majority (62%) have crossed
    - Horizon choice is presentationally consequential
3. **The 2-year consecutive rule has moderate effect**
    - 5.7pp reduction vs 1-year
    - 2-of-3 provides minimal relaxation (0.2pp)
    - This suggests failures are not “just missed” but substantial
4. **Stochastic elements affect success rate, not median timing**
    - Deterministic: 2039
    - Stochastic median: 2039
    - Shocks create “failures” more than “delays”

What this does NOT show

1. **“80% of conservatism”** — Not a valid claim
2. **“Aggregation doesn’t matter”** — Only true within tested range; softmin matters
3. **“2-of-3 is equivalent to 1-year”** — It is equivalent to 2-year, not 1-year
4. **“Shocks have no effect”** — They have large effect on success rate, not median

---

Implications for Model Assessment

The core question is now clearly: floor justification

The ablation shows that: - Floors dominate all other structural choices - Floor level (60 vs 55 vs 65) has enormous effect - Floor uniformity (same for M/R/N/G) is untested

Therefore: The substantive debate should focus on: 1. Why 60 specifically? 2. Why uniform across M/R/N/G? 3. Is 60 a capability threshold or a convenience choice? 4. What empirical/theoretical basis supports these levels?

Reporting must change

Current: > “91% never crosses by 2040”

Should be: > “9% cross by 2040; 37% cross by 2045; 62% cross by 2050”

The “intuition gap” is now clearer

If intuitive expectation is 2033-2035: - Deterministic path: 2039 (close) - Stochastic median: 2039 (close) - Main divergence: floor failures (38pp) + horizon truncation

The gap is not “model is too conservative” but “floors may be too strict relative to intuitive readiness thresholds.”

Recommended Next Steps (No Model Changes Yet)

1. **Document floor justification**
    - Current: expert judgment, no empirical basis
    - Need: conceptual framework for why 60, why uniform
2. **Report multi-horizon probabilities**
    - Always show 2040, 2045, 2050
    - Never say “never” without horizon qualifier
3. **Test floor variations as audit, not tuning**
    - Uniform 55, 60, 65
    - Differential (G=55, N=55, M=R=60)
    - Soft floors / penalty functions
    - Report sensitivity, don’t change defaults yet
4. **Validate consecutive-years results**
    - ✓ Done: confirmed 2-of-3 ≈ 2-year (not 1-year)

Questions for Floor Justification

To properly justify floors=60, should be able to answer:

1. **What concrete capability does Memory=60 represent?**
    - Persistent context across X sessions?
    - Y% reliability of state reconstruction?
2. **Why is Governance=60 the same as Memory=60?**
    - Are these comparable scales?
    - Should institutional readiness have different threshold?
3. **What happens at 59 vs 60?**
    - Is there a phase transition?
    - Or is this continuous with arbitrary cutoff?
4. **Empirical basis:**
    - Any historical systems that achieved “society-like” function at these levels?
    - Or is this aspirational specification?

Without answers, floors=60 remains an unexamined assumption that happens to dominate model output.

Status: Validation complete. Ready for floor justification audit.

Model V2 Design: Two-Stage AI Agent Network Forecast

Status: Design document for next-generation forecasting model
Objective: Separate “first emergence” from “broad viability” to resolve intuition-model conflict
Date: 2026-03-25

---

1. Core Insight: Two Different Events

The intuition-model conflict arises from conflating two distinct phenomena:

Aspect	Event A: First Emergence	Event B: Broad Viability
What	First real Level-3-like network appears	Mature, trusted, routine deployment
Scope	Bounded, limited domains	Broad, cross-vendor, general use
Governance	Guardrails present, human oversight	Strong governance, lower oversight
Reliability	“Good enough” for specific use	High reliability, institutional trust
When (intuition)	2030-2035	2036-2040+
Current model	Not explicitly modeled	What current model forecasts

Key realization: The current model forecasts Event B (mature deployability), while intuition often targets Event A (first emergence).

---

2. Event Definitions

Event A: First Emergence (Bounded-Scope Level-3)

Definition: The first year when a network of AI agents demonstrates: - Autonomous task completion in bounded domains - Meaningful multi-agent coordination - Commercial or operational viability (not just research demo) - Some governance (not necessarily mature) - Limited scope: specific use cases, controlled environment

Characteristics: - Soft feasibility: pillars can be “good enough” rather than “excellent” - Capabilities-weighted: C, E, D matter most - Governance-lite: G exists but is immature - Early interoperability: N works for specific partners/protocols

Examples: - Enterprise internal agent networks (2028-2032) - Vertical-specific agent ecosystems (finance, coding) (2030-2034) - Closed beta multi-agent platforms (2031-2035)

Event B: Broad Viability (Mature Deployability)

Definition: The year when agent networks become: - Cross-vendor interoperable - Routinely trusted for important decisions - Governed by established frameworks - Deployable with manageable risk - Institutionally

accepted

Characteristics: - Harder requirements: all pillars must be mature - Governance-heavy: G, R are critical - Network effects: N must be universal, not fragmented - Institutional trust: requires track record

Examples: - Consumer-facing agent social networks (2036-2042) - Autonomous economic agents with minimal oversight (2038-2045) - Cross-platform agent standards widely adopted (2037-2043)

---

3. Architectural Changes: V1 → V2

3.1 From Hard Floors to Soft Feasibility

V1 (Current):

```
if pillar < floor:
    fail
```

V2 (Proposed):

```
feasibility = sigmoid((pillar - threshold) / steepness)
```

Rationale: - V1 treats 59 as complete failure, 60 as success - V2 treats feasibility as continuous, with steep transition around threshold - Event A uses shallower slope (more forgiving) - Event B uses steeper slope (stricter)

3.2 From Independent Pillars to Coupled System

V1 (Current): - Each pillar evolves independently - No interaction between C and N, or G and D

V2 (Proposed): - Endogenous coupling between pillars

Coupling mechanisms:

From	To	Mechanism	Rationale
C (Capability)	D (Demand)	Better capabilities → more use cases → higher demand	Capability creates demand
E (Efficiency)	D (Demand)	Lower costs → broader adoption	Economic accessibility
D (Demand)	N (Network)	More users → need better coordination protocols	Scale requires standards
N (Network)	R (Reliability)	Better coordination → fewer failures	Infrastructure stability
G (Governance)	Deployment Scope	Better governance → broader allowed deployment	Regulatory acceptance
D (Demand)	G (Governance)	More deployment → pressure for governance	Regulatory response

Mathematical form:

```
growth_mu_D(t) = base_D + α×C(t) + β×E(t)
growth_mu_N(t) = base_N + γ×D(t-1)
shock_G(t) = f(D(t), regulatory_events)
```

3.3 From Static Growth to Regime-Switching

V1 (Current): - Fixed growth rates per scenario - Random shocks but no structural shifts

V2 (Proposed): - Regime-switching for Governance (G) and Network (N)

Governance regimes: 1. **Wild West:** Minimal regulation, fast innovation, high risk 2. **Crisis Response:** Major incident triggers regulatory action 3. **Framework Emergence:** Standards and guidelines developed 4. **Mature Governance:** Established oversight, predictable rules

Network regimes: 1. **Fragmented:** Proprietary protocols, siloed systems 2. **Consolidation:** Winners emerge, some interoperability 3. **Standardization:** Industry standards adopted 4. **Universal:** Seamless cross-platform coordination

Transition dynamics: - Triggered by threshold crossings (e.g., D > 70 triggers governance response) - History-dependent (once in regime 3, hard to go back to 1) - Stochastic transition probabilities

3.4 From Binary Crossing to Time-to-Event Distribution

V1 (Current):

```
crossing = True if (index > 75 AND floors_met) for 2 years
result: "crosses in 2038" or "never"
```

V2 (Proposed):

```
hazard_A(t) = f(C(t), E(t), D(t), soft_feasibility(G,N,R,M))
hazard_B(t) = f(all_pillars, hard_feasibility(G,N,R,M), coupling_effects)

result: "P(Event A in 2032) = 25%, P(Event A by 2035) = 60%"
        "P(Event B in 2038) = 15%, P(Event B by 2042) = 55%"
```

Reporting: - Survival curves: “P(not yet emerged by year t)” - Hazard rates: “P(emergence in year t | not yet emerged)” - Cumulative distributions: “P(emerged by year t)”

---

4. Mathematical Specification

4.1 Soft Feasibility Function

For each pillar i, define feasibility:

```
φ_i(y_i) = 1 / (1 + exp(-k_i × (y_i - θ_i)))

where:
  y_i = pillar score (0-100)
```

```
  θ_i = threshold (inflection point)
  k_i = steepness (higher = more threshold-like)
```

Event A parameters (forgiving): - k = 0.1 (gradual transition) - θ = 50 (lower threshold)

Event B parameters (strict): - k = 0.2 (sharper transition) - θ = 60 (higher threshold)

4.2 Coupled Growth Dynamics

Base growth with coupling:

```
Δz_C(t) = g_C + ε_C(t)                          # Capability: exogenous
Δz_E(t) = g_E + ε_E(t)                          # Efficiency: exogenous
Δz_D(t) = g_D + α×z_C(t) + β×z_E(t) + ε_D(t)    # Demand: driven by C, E
Δz_N(t) = g_N + γ×z_D(t-1) + ε_N(t)             # Network: driven by D
Δz_R(t) = g_R + δ×z_N(t) + ε_R(t)               # Reliability: driven by N
Δz_M(t) = g_M + ε_M(t)                          # Memory: exogenous

z_G(t) = regime-dependent:
  Regime 1: g_G = low, high volatility
  Regime 2: g_G = medium, crisis shocks possible
  Regime 3: g_G = medium, lower volatility
  Regime 4: g_G = high, low volatility
```

Regime transitions:

```
P(Regime(t+1) = r' | Regime(t) = r, z(t)) = f(z(t), threshold_matrix)

Example:
  If z_D > 70 AND z_G < 40: high P(transition to Regime 2 - crisis response)
  If z_G > 60 for 3 years: high P(transition to Regime 4 - mature)
```

4.3 Hazard Functions

Event A (First Emergence):

```
h_A(t) = h0_A × exp(β_C×z_C + β_E×z_E + β_D×z_D)
         × φ_G(z_G) × φ_N(z_N) × φ_R(z_R) × φ_M(z_M)

Interpretation:
- Exponential in core capabilities (C, E, D)
- Modulated by soft feasibility of support pillars (G, N, R, M)
- High hazard when C,E,D strong, even if G,N,R,M mediocre
```

Event B (Broad Viability):

```
h_B(t) = h0_B × exp(β_all × weighted_avg(z))
         × min(φ_G(z_G), φ_N(z_N), φ_R(z_R), φ_M(z_M))^γ

Interpretation:
- Based on all pillars
- Strongly penalized by weakest critical pillar (min function)
- γ > 1 makes this a "bottleneck-sensitive" hazard
```

4.4 Survival and Cumulative Distribution

```
S(t) = P(T > t) = exp(-∫_0^t h(s) ds)          # Survival: not yet happened
F(t) = 1 - S(t) = P(T ≤ t)                     # CDF: happened by t
f(t) = h(t) × S(t)                             # PDF: happens at t
```

Reporting: - Median time: t where F(t) = 0.5 - 90% CI: [t where F(t) = 0.05, t where F(t) = 0.95] - P(by year): F(2030), F(2035), F(2040), etc.

---

5. Parameterization Strategy

5.1 Exogenous Parameters (from data/expert)

Parameter	Source	Confidence
C, E growth rates	Stanford HAI, Epoch trends	Medium
M growth rate	Expert judgment	Low
Initial values (2026)	Expert judgment	Low
Volatilities	Expert judgment	Low

5.2 Coupling Parameters (to be estimated/calibrated)

Parameter	How to Set	Prior
α (C→D)	Historical analogies: did better AI increase demand?	0.3
β (E→D)	Elasticity: demand response to cost decreases	0.4
γ (D→N)	Network effects literature	0.2
δ (N→R)	Infrastructure reliability studies	0.25

Validation: Sensitivity analysis on coupling strength.

5.3 Regime Parameters (expert elicitation needed)

Parameter	Question for Experts
Transition thresholds	At what demand level does governance pressure spike?
Crisis probabilities	How likely is a major AI incident triggering regulation?
Recovery rates	How fast can governance mature after crisis?

5.4 Event Definition Parameters (normative choice)

Parameter	Event A	Event B
Soft threshold θ	50	60

Parameter	Event A	Event B
Steepness k	0.1 (gradual)	0.2 (sharp)
Weakest-link exponent γ	1.0 (moderate)	2.0 (strong)

---

6. Data Sources for Calibration

6.1 Direct Proxies

Pillar	Data Source	Notes
C (Capability)	Epoch Capabilities Index (ECI)	Quarterly updates
	MMLU, HumanEval, SWE-bench	Specific benchmarks
E (Efficiency)	AI training cost trends	Stanford HAI
	Inference cost per token	OpenAI, Anthropic pricing
D (Demand)	AI investment flows	Crunchbase, PwC
	Enterprise adoption surveys	McKinsey, Deloitte

6.2 Governance Proxies

Proxy	Source	Mapping to G
National AI strategies	OECD.AI	G framework presence
AI regulations enacted	Stanford HAI	G maturity
AI incidents database	AIAAIC	G failure rate
Governance frameworks	NIST, ISO 42001	G adoption

6.3 Network Proxies

Proxy	Source	Mapping to N
Multi-agent platform adoption	GitHub, npm	N usage
Protocol standardization	IETF, W3C	N standards
Vendor interoperability	Industry reports	N maturity

6.4 Calibration Approach

1. **Backcast:** Fit 2020-2025 dynamics to observed trends
2. **Validate:** Out-of-sample test 2023-2025 if data allows
3. **Project:** Forward simulation with uncertainty
4. **Update:** Quarterly refresh as new data arrives

---

7. V1 vs V2 Comparison

Aspect	Model V1 (Current)	Model V2 (Proposed)
Events	Single “crossing”	Two: First Emergence + Broad Viability
Floors	Hard (59=fail, 60=pass)	Soft (sigmoid feasibility)
Pillars	Independent	Coupled (C,E→D→N→R)
Governance	Static growth	Regime-switching
Output	Binary: year or “never”	Distribution: survival curves
Intuition fit	Poor (2038 vs 2033 gut)	Good (separate events)
Complexity	Medium	High
Validation	Limited	More tractable (time-to-event)
Uncertainty	Hidden in “never”	Explicit in distributions

---

8. Implementation Roadmap

Phase 1: Soft Feasibility (2-3 days)

- Replace hard floors with sigmoid functions
- Keep rest of model unchanged
- Test sensitivity to k and θ

Phase 2: Two Events (3-4 days)

- Implement Event A (emergence) with forgiving parameters
- Implement Event B (viability) with strict parameters
- Report both survival curves

Phase 3: Basic Coupling (4-5 days)

- Add C→D and E→D coupling
- Add D→N coupling
- Calibrate coupling strengths

Phase 4: Regime Switching (5-7 days)

- Implement G regimes
- Implement N regimes
- Define transition rules

Phase 5: Validation & Documentation (3-4 days)

- Backcast to 2020-2025
- Sensitivity analysis
- Full documentation

Total: 3-4 weeks for full V2

---

9. What Remains Uncalibrated / Expert Judgment

Inherently Uncertain (no data can resolve)

1. **What is "good enough" governance?**
   - Even with frameworks, actual performance varies
   - Cultural differences in governance acceptance
2. **When does fragmentation become interoperability?**
   - Network effects are path-dependent
   - History of tech standards shows unpredictability
3. **Crisis timing and severity**
   - Major AI incidents are rare (thankfully)
   - Cannot statistically predict next crisis

Addressable with Better Data

1. **Coupling strengths**
   - Historical analogies: internet, mobile, cloud adoption
   - Can narrow ranges with more research
2. **Regime transition probabilities**
   - Comparative analysis of other regulated technologies
   - Nuclear, aviation, financial systems as analogies
3. **Initial conditions**
   - Better measurement of current governance maturity
   - More systematic capability assessment

---

10. Honest Assessment: What V2 Achieves and Doesn't

What V2 Achieves

✅ Separates intuition (first emergence) from model (broad viability)
✅ Makes uncertainty explicit in distributions, not binary "never"
✅ Captures key intuitions: coupling, regime shifts, soft thresholds
✅ More realistic microfoundations for governance and network dynamics
✅ Better validation potential via time-to-event methods

What V2 Doesn't Solve

❌ Still requires expert judgment for many parameters
❌ Still no direct historical data on "agent societies"
❌ Still sensitive to normative choices (what counts as "emergence")
❌ Cannot predict black swan events or breakthroughs
❌ Does not reduce fundamental uncertainty, only structures it better

The Core Trade-off

Approach	Simplicity	Realism	Validation	Communication
V1 (Current)	✓✓✓	✓	✓	×
V2 (Proposed)	✓	✓✓✓	✓✓	✓✓

V2 is more complex but more honest about what we know and don't know.

---

11. Immediate Next Steps (No Code Yet)

Before implementing V2:

1. **Validate the conceptual framework**
    - Does "first emergence vs broad viability" resonate?
    - Are the coupling mechanisms plausible?
2. **Gather calibration data**
    - Collect historical proxy time series
    - Identify expert elicitation candidates
3. **Decide on soft floor parameters**
    - What θ and k for Event A vs B?
    - Sensitivity to these choices?
4. **Define regimes precisely**
    - What are the measurable criteria for each G regime?
    - What triggers transitions?

Status: Design complete. Ready for conceptual validation before implementation.

Model V2 Formal Specification

Status: Mathematical specification (NOT implementation)
Purpose: Exact model definition before any coding
Constraint: NO illustrative outputs - only structure, parameters, and calibration strategy
Date: 2026-03-25

---

1. Event Definitions (Exact)

1.1 Event A: First Emergence

Formal Definition:
> The first calendar year in which at least one commercially deployed network of AI agents demonstrates: > 1. Autonomous task completion in production (not research/demo) for bounded use cases > 2. Multi-agent coordination without continuous human micromanagement
> 3. Commercial or operational viability (generates value, not just novelty) > 4. Deployment under strong local controls, limited blast radius, and human oversight

What COUNTS as Event A: - Enterprise-internal agent networks handling >$10M/year of operational decisions (2030-2033) - Vertical-specific platforms (coding, trading, customer service) with >1000 paying customers (2031-2034) - Closed-beta multi-agent systems with real economic transactions (2032-2035) - Government/institutional deployments in controlled domains (healthcare, logistics) (2033-2036)

What does NOT count as Event A: - Research demos (AutoGPT, babyagi) - Single-agent systems (not networks) - Human-in-the-loop systems requiring continuous supervision - Systems with no economic value creation - Systems that fail in production within 6 months

Governance requirement for Event A:
> Governance need not be broadly institutionalized or cross-jurisdictional. Bounded-scope deployment must be possible under: > - Strong local controls within the deploying organization > - Human oversight at strategic/decision level (not operational) > - Limited blast radius (failure contained to specific use case) > - Audit trails exist but need not be standardized

1.2 Event B: Broad Viability

Formal Definition:
> The first calendar year when agent networks satisfy ALL of: > 1. Cross-vendor interoperability (agents from different providers coordinate) > 2. Routine deployment with manageable risk (not experimental) > 3. Institutional trust (regulated industries deploy without special waivers) > 4. Governance frameworks mature and widely adopted > 5. Meaningful deployment with reduced human oversight (operational, not strategic)

What COUNTS as Event B: - Consumer-facing agent social networks with >10M users (2036-2042) - Cross-platform agent standards (IEEE, W3C, or de facto) adopted by >3 major vendors (2037-2043) - Autonomous economic agents operating in regulated markets without special exemptions (2038-2045) - Government services delivered primarily by agent networks (2039-2047)

What does NOT count as Event B: - Single-vendor walled gardens - Experimental deployments with extensive human supervision - Systems requiring regulatory exemptions or sandboxes - Niche use cases without broad applicability

Governance requirement for Event B:
> Governance MUST be: > - Institutionalized across jurisdictions (not just one organization's policy) > - Standardized (common frameworks, not ad-hoc) > - Enforced (violations have consequences) > - Proven (track record of handling incidents)

---

2. Mathematical Specification (Exact)

2.1 Time Structure

- Discrete time steps: $t \in \{2026, 2027, \ldots, 2070\}$
- Annual resolution (no sub-year steps)
- Horizon: 2026-2070 (45 years, right-censoring at 2070)

2.2 State Space

Observed variables (7 pillars): - $C(t), E(t), M(t), R(t), N(t), G(t), D(t) \in [0, 100]$

Latent variables: - $z_C(t), \ldots, z_D(t) \in \mathbb{R}$ (logit scale for each pillar) - $Regime_G(t) \in \{1, 2, 3, 4\}$ (governance regime) - $Regime_N(t) \in \{1, 2, 3, 4\}$ (network regime)

2.3 Observation Model (Logit Transform)

```
y_i(t) = 100 × σ(z_i(t))

where σ(x) = 1 / (1 + exp(-x)) is the sigmoid

Inverse:
z_i(t) = logit(y_i(t)/100) = log(y_i(t) / (100 - y_i(t)))
```

Properties: - Maps [0, 100] to [-∞, +∞] - Natural saturation: $y \to 100$ as $z \to +\infty$ - Linear dynamics on z-scale approximate logistic growth on y-scale

2.4 State Evolution (Coupled Local Linear Trend)

Core pillars (exogenous):

```
z_C(t) = z_C(t-1) + g_C(t-1) + ε_C(t)
z_E(t) = z_E(t-1) + g_E(t-1) + ε_E(t)
z_M(t) = z_M(t-1) + g_M(t-1) + ε_M(t)

where:
  g_i(t) = persistence_i × g_i(t-1) + (1 - persistence_i) × ḡ_i + η_i(t)
  ε_i(t) ~ N(0, σ_level_i)  [level shock]
  η_i(t) ~ N(0, σ_growth_i)  [growth rate shock]
```

Coupled pillars (endogenous):

```
z_D(t) = z_D(t-1) + g_D(t-1) + α_C×z_C(t-1) + α_E×z_E(t-1) + ε_D(t)
z_N(t) = z_N(t-1) + g_N(t-1) + β_D×D(t-1) + regime_effect_N(t) + ε_N(t)
z_R(t) = z_R(t-1) + g_R(t-1) + γ_N×N(t-1) + ε_R(t)

where:
  D(t-1) = 100 × σ(z_D(t-1))  [back to observation scale]
  regime_effect_N(t) = f(Regime_N(t))
```

Governance (regime-switching):

```
z_G(t) = z_G(t-1) + g_G(t-1) + regime_effect_G(t) + ε_G(t)

Regime_G(t) | Regime_G(t-1), D(t-1), shocks
  Transition matrix depends on demand level and incident history
```

2.5 Regime Definitions (Governance)

Regime	Name	Characteristics	Transition triggers
1	Wild West	Minimal regulation, fast innovation, high risk	High D + low G triggers → 2
2	Crisis Response	Major incident occurred, reactive regulation	After major incident
3	Framework Emergence	Standards developing, proactive governance	G sustained > 40 for 3 years → 4
4	Mature Governance	Established oversight, predictable rules	(absorbing or can degrade)

Regime effects on growth:

```
regime_effect_G(t) =
  +0.05 if Regime_G(t) = 4 (mature governance enables faster deployment)
   0.00 if Regime_G(t) = 3
  -0.10 if Regime_G(t) = 2 (crisis slows things down)
  +0.10 if Regime_G(t) = 1 (wild west allows fast but risky growth)
```

2.6 Soft Feasibility Function

For each pillar i, define feasibility φ_i:

```
φ_i(y_i; θ_i, k_i) = 1 / (1 + exp(-k_i × (y_i - θ_i) / 10))

where:
  y_i = pillar score on [0, 100]
  θ_i = inflection point (where feasibility = 0.5)
  k_i = steepness parameter (higher = sharper threshold)
  Division by 10 scales to roughly [-10, +10] range for typical y differences
```

Event A parameters: | Pillar | θ_i | k_i | Rationale | |———|——|——|————| | C | 45 | 0.15 | Capability matters most, lower threshold | | E | 45 | 0.15 | Efficiency enables deployment | | D | 40 | 0.12 | Demand can be emerging | | M | 50 | 0.10 | Memory should be decent | | R | 50 | 0.10 | Reliability important but bounded-scope forgiving | | N | 45 | 0.10 | Early interoperability okay | | G | 40 | 0.08 | Governance can be immature (local controls suffice) |

Event B parameters: | Pillar | θ_i | k_i | Rationale | |———|——|——|————| | C | 60 | 0.20 | High capability required | | E | 60 | 0.20 | Efficiency at scale | | D | 55 | 0.18 | Established demand | | M | 60 | 0.18 | Robust memory | | R | 65 | 0.22 | High reliability critical | | N | 65 | 0.22 | Universal interoperability | | G | 65 | 0.25 | Mature governance essential (steepest) |

2.7 Hazard Functions (Time-to-Event)

Event A Hazard:

```
h_A(t) = h0_A × exp(λ_C×z_C(t) + λ_E×z_E(t) + λ_D×z_D(t))
         × φ_G(y_G(t)) × φ_N(y_N(t)) × φ_R(y_R(t)) × φ_M(y_M(t))

where:
  h0_A = baseline hazard (to be calibrated)
  λ_C, λ_E, λ_D = loadings on core capabilities
  φ_G, φ_N, φ_R, φ_M = soft feasibility of support pillars
```

Properties: - Exponential in core capabilities (C, E, D) - Modulated by feasibility of support pillars - Even with weak G/N/R/M, high C/E/D can drive hazard

Event B Hazard:

```
h_B(t) = h0_B × exp(Σ_i λ_i×z_i(t))
         × [min(φ_G(y_G(t)), φ_N(y_N(t)), φ_R(y_R(t)), φ_M(y_M(t)))]^γ

where:
  h0_B = baseline hazard (lower than h0_A)
  γ = 2.0 (bottleneck sensitivity)
  min() = bottleneck function
```

Properties: - Based on all pillars - Strongly penalized by weakest critical pillar - γ > 1 creates severe bottleneck effect - Governance and network are hardest to satisfy

2.8 Survival and Cumulative Distribution

```
S_A(t) = P(T_A > t) = exp(-Σ_{s=2026}^t h_A(s))
F_A(t) = P(T_A ≤ t) = 1 - S_A(t)

S_B(t) = P(T_B > t) = exp(-Σ_{s=2026}^t h_B(s))
F_B(t) = P(T_B ≤ t) = 1 - S_B(t)
```

Reporting quantities: - Median: t where F(t) = 0.5 - 90% CI: [t_0.05, t_0.95] where F(t_p) = p - P(by year): F(2030), F(2035), F(2040), etc.

---

3. Parameter Strategy

3.1 Parameter Classification

Parameter	Type	Source	Confidence
Initial z_i(2026)	Point estimate	Expert judgment + proxy data	LOW
ḡ_i (long-run growth)	Point estimate	Stanford HAI, Epoch trends	MEDIUM
persistence_i	Point estimate	Literature on technology diffusion	LOW
σ_level_i, σ_growth_i	Distribution	Expert elicitation	LOW
α_C, α_E (coupling to D)	Range [0.1, 0.5]	Historical analogies	MEDIUM
β_D (coupling to N)	Range [0.1, 0.3]	Network effects literature	MEDIUM
γ_N (coupling to R)	Range [0.1, 0.3]	Infrastructure reliability	LOW
θ_i (thresholds)	Range	Maturity framework mapping	MEDIUM
k_i (steepness)	Distribution	Expert elicitation on “fuzziness”	LOW
h0_A, h0_B	Calibrated	Match historical analogies	MEDIUM
λ_i (hazard loadings)	Normalized	Equal weighting as starting point	LOW
γ (bottleneck exponent)	Range [1.5, 3.0]	Sensitivity analysis	LOW

3.2 Sensitivity Strategy

Must vary: - All θ_i ± 10 points - All k_i ± 50% - Coupling parameters across their ranges - Baseline hazards h0_A, h0_B over 2 orders of magnitude - Regime transition probabilities

Output: Uncertainty intervals, not point estimates.

---

4. Calibration Plan

4.1 Direct Calibration (where possible)

Target	Data Source	How to Use
C growth	Epoch ECI, MMLU, HumanEval	Fit trend 2020-2025, project
E growth	Stanford HAI cost trends	Extrapolate cost decline
D growth	AI investment, enterprise adoption	Elasticity to C, E

Target	Data Source	How to Use
M, R, G, N	No direct data	Expert elicitation only

4.2 Proxy Calibration

Pillar	Proxy	Source
G (Governance)	Oxford Gov AI Readiness, OECD AI Policy	Annual indices
	AI incident rates	AIAAIC database
	Regulatory enactment	Stanford HAI, national trackers
N (Network)	Multi-agent repos (GitHub)	Dependency graphs
	Protocol standardization	IETF, W3C activity
	Cross-platform tools	npm, PyPI metrics

4.3 Historical Analogies for Baseline Hazard

Event A analogies: - First cloud deployments (AWS 2006): time from concept to production viability - First mobile apps (App Store 2008): time from SDK to commercial viability - First SaaS (Salesforce 1999): time from idea to enterprise adoption

Event B analogies: - Cloud maturity (AWS → multi-cloud standardization): 2010-2020 - Mobile maturity (iOS → cross-platform): 2008-2015 - Internet maturity (ARPANET → commercial internet): 1983-1995

Use: Calibrate h0 to match analogous historical durations.

4.4 Expert Elicitation Protocol

For parameters without data: 1. Define observable implications of each parameter value 2. Ask experts for P(parameter < x) distributions 3. Aggregate using Cooke's Classical Model (weighted by calibration) 4. Propagate uncertainty through Monte Carlo

Parameters needing elicitation: - θ_G, θ_N (what's "good enough" governance/network?) - k_G, k_N (how sharp is the feasibility transition?) - σ_G, σ_N (volatility of institutional factors) - Regime transition probabilities

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5. Validation Plan

5.1 Sanity Checks

Check	Expected	If Fails
C, E fastest growing	Yes	Check growth rate priors
G slowest growing	Yes	Consistent with institutional friction
Event A before Event B	Yes	By construction (higher hazard)

Check	Expected	If Fails
Both events after 2025	Yes	No backcasting failure
S(t) decreasing	Yes	Mathematical check

5.2 Ablations (Required)

1. **No coupling:** Set α, β, γ = 0
2. **No regimes:** Fix Regime_G = 3, Regime_N = 2
3. **Hard floors:** Replace sigmoid with step function
4. **Single event:** Only Event A or only Event B
5. **Exogenous only:** No D→N→R coupling
6. **Softer/steeper feasibility:** Vary all k_i ± 50%

5.3 Pseudo-Backtests

Test: Can model "predict" historical technology diffusion?

Approach: - Treat cloud, mobile, internet as "Event A" analogies - Fit model to 2000-2020 data - See if it predicts 2020-2025 correctly - Use to validate coupling strengths

Limitation: No true "Level-3 agent" historical data exists.

5.4 Alternative Event Definitions

Test robustness: - Event A with θ_G = 35 vs 45 vs 55 - Event B with γ = 1.5 vs 2.0 vs 3.0 - Different bottleneck definitions (min vs geometric mean)

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6. Risks and Open Questions

6.1 Fundamental Uncertainties

1. **No historical precedent for Level-3 agents**
   - All analogies (cloud, mobile) are imperfect
   - May be category error to assume continuity
2. **Discontinuity risk**
   - Model assumes smooth dynamics
   - Breakthrough (GPT-4 moment for agents) not captured
3. **Regime transitions are speculative**
   - We have no validated theory of AI governance evolution
   - Crisis timing inherently unpredictable
4. **Coupling parameters unobserved**
   - Cannot directly measure "C → D elasticity"
   - Historical analogies may not apply

6.2 Model Limitations

1. **Annual resolution misses sub-year dynamics**
   - If emergence happens mid-year, model rounds
2. **No spatial/geographic differentiation**

    - Governance regimes vary by jurisdiction
    - Model treats world as homogeneous
3. **Binary event definition**
    - Real world has degrees of emergence/viability
    - Model forces discrete threshold
4. **No feedback from forecast to reality**
    - Model is passive observer
    - Real forecasts can accelerate/slow development

6.3 Ethical/Strategic Considerations

1. **Self-fulfilling prophecy risk**
    - If forecast says “2038”, does this slow investment?
    - Or accelerate if perceived as race?
2. **Strategic misrepresentation**
    - Incentives to forecast early (hype) or late (caution)
    - Model must resist manipulation
3. **Distributional impacts**
    - Who benefits from early emergence? Late?
    - Model is agnostic but stakes are high

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7. Deliverables and Next Steps

7.1 Before Implementation

- ☐ Validate event definitions with stakeholders
- ☐ Confirm mathematical framework choice (discrete-time logit hazard)
- ☐ Finalize parameter ranges (not point values)
- ☐ Design expert elicitation protocol
- ☐ Identify historical analogy datasets

7.2 Implementation Phases

Phase	Duration	Deliverable
1: Core structure	1 week	Basic hazard functions, no coupling
2: Coupling	1 week	C,E→D→N→R dynamics
3: Regimes	1 week	G and N regime switching
4: Calibration	2 weeks	Fit to proxies, expert elicitation
5: Validation	1 week	Ablations, sensitivity, documentation

Total: 6 weeks

7.3 Success Criteria

Model V2 is successful if: 1. It produces interpretable survival curves (not binary “cross/don’t cross”) 2. Event A and Event B have meaningfully different distributions 3. Uncertainty is explicit in output (ranges, not points) 4.

Sensitivity to floor/governance parameters is clearly documented 5. It can be updated quarterly as new data arrives

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8. Explicit Absence

NOT in this specification: - ❌ Any claimed median year (2033, 2038, etc.) - ❌ Any claimed probability ("60% by 2035") - ❌ Any "illustrative" outputs - ❌ Implementation code - ❌ Parameter point values (only ranges/priors)

What IS in this specification: - ✅ Exact mathematical forms - ✅ Exact event definitions with examples - ✅ Parameter classification and sources - ✅ Calibration strategy - ✅ Validation plan - ✅ Honest uncertainty documentation

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Status: Specification complete. Awaiting validation before implementation.